

Recognition and Cleavage of Human tRNA Methyltransferase TRMT1 by the SARS-CoV-2 Main Protease

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eLife Assessment

This manuscript provides **important** structural insights into the recognition and degradation of the host tRNA methyltransferase TRMT1 by SARS-CoV-2 protease nsp5 (Mpro). The data provide **compelling** support for the main conclusions of the authors. These results will be of interest to researchers studying structures, substrate recognition and specificity of viral proteases and their action on cellular targets.

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Abstract

The SARS-CoV-2 main protease (M^{Pro}, or Nsp5) is critical for the production of functional viral proteins during infection and, like many viral proteases, can also target host proteins to subvert their cellular functions. Here, we show that the human tRNA methyltransferase TRMT1 can be recognized and cleaved by SARS-CoV-2 M^{Pro}. TRMT1 installs the *N*²,*N*²-dimethylguanosine (m2,2G) modification on mammalian tRNAs, which promotes global protein synthesis and cellular redox homeostasis. We find that M^{Pro} can cleave endogenous TRMT1 in human cell lysate, resulting in removal of the TRMT1 zinc finger domain. TRMT1 proteolysis results in elimination of TRMT1 tRNA methyltransferase activity and reduced tRNA binding affinity. Evolutionary analysis shows that the TRMT1 cleavage site is highly conserved in mammals, except in Muroidea, where TRMT1 is likely resistant to cleavage. In primates, regions outside the cleavage site with rapid evolution could indicate adaptation to ancient viral pathogens. Furthermore, we determined the structure of a TRMT1 peptide in complex with M^{Pro}, revealing a substrate binding conformation distinct from the majority of available M^{Pro}-peptide complexes. Kinetic parameters for peptide cleavage show that the TRMT1(526-536) sequence is cleaved with comparable efficiency to the M^{Pro}-targeted nsp8/9 viral cleavage site. Mutagenesis studies and molecular dynamics simulations together indicate that kinetic discrimination occurs during a later step of M^{Pro}-mediated proteolysis that follows substrate binding. Our results provide new information about the structural

basis for M^{Pro} substrate recognition and cleavage, the functional roles of the TRMT1 zinc finger domain in tRNA binding and modification, and the regulation of TRMT1 activity by SARS-CoV-2 M^{Pro}. These studies could inform future therapeutic design targeting M^{Pro} and raise the possibility that proteolysis of human TRMT1 during SARS-CoV-2 infection suppresses protein translation and oxidative stress response to impact viral pathogenesis.

Significance Statement

Viral proteases can strategically target human proteins to manipulate host biochemistry during infection. Here, we show that the SARS-CoV-2 main protease (M^{Pro}) can specifically recognize and cleave the human tRNA methyltransferase enzyme TRMT1, and that cleavage of TRMT1 cripples its ability to install a key modification on human tRNAs that is critical for protein translation. Our structural and functional analysis of the M^{Pro}-TRMT1 interaction shows how the flexible M^{Pro} active site engages a conserved sequence in TRMT1 in an uncommon binding mode to catalyze its cleavage and inactivation. These studies provide new insights into substrate recognition by SARS-CoV-2 M^{Pro} that could help guide future antiviral therapeutic development and show how proteolysis of TRMT1 during SARS-CoV-2 infection impairs both TRMT1 tRNA binding and tRNA modification activity to disrupt host translation and potentially impact COVID-19 pathogenesis or phenotypes.

Introduction

SARS-CoV-2 has resulted in over 6 million deaths worldwide since the start of the COVID-19 pandemic in early 2020 (1, 2). The development of mRNA and other vaccines has played a large and critical role in reducing mortality since their introduction in 2021 (3–6), but a fundamental understanding of coronavirus host-protein interactions and biology continues to be an important goal to inform ongoing and future therapeutic design. The SARS-CoV-2 main protease (M^{Pro}, or Nsp5) is a well-studied antiviral drug target because its activity is essential for viral replication. More than 600 M^{Pro} crystal structures – the majority with bound small molecule inhibitors – are currently available in the protein databank (7). M^{Pro} is necessary for the proteolysis of 11 different cleavage sites in the two SARS-CoV-2 extended viral polypeptides, which results in the liberation of mature non-structural proteins (nsps) that are essential for host infection and viral propagation (8–13). Approximately twenty structures of M^{Pro} in complex with different viral cleavage sequences have been determined (14–16), in which the M^{Pro} homodimer has a peptide substrate bound to its active site (Figure 1A). However, a detailed understanding of SARS-CoV-2 M^{Pro} substrate selectivity and cleavage efficiency remains poorly defined, and the structural characterization of interactions between the viral protease and human host protein targets remains underexplored (17).

Early in the COVID-19 pandemic, Gordon *et al.* expressed tagged SARS-CoV-2 proteins in human embryonic kidney cells (HEK293T/17) and employed affinity purification mass spectrometry (AP-MS) to map proteome-wide virus-host protein interactions, in which a putative interaction between catalytically inactive SARS-CoV-2 M^{Pro} (Cys145Ala) and the human tRNA methyltransferase TRMT1 was identified (18). Analogous AP-MS experiments with wild-type M^{Pro} found no stable interaction with TRMT1, and subsequent mapping with SARS-CoV-1 proteins also found a TRMT1 interaction exclusively with Cys145Ala M^{Pro}, suggesting that coronavirus M^{Pro} may recognize and actively cleave human TRMT1 in cells (18, 19). TRMT1 is a tRNA-modifying enzyme responsible for installing N²,N²-dimethylguanosine (m²,2G), an abundant tRNA modification found at the G26 position of human tRNAs (20). Human HEK293T cells lacking TRMT1, and therefore the m²,2G26 modification, have significantly decreased global protein synthesis and reduced proliferation (21). Human neural stem cells with TRMT1 knockdowns

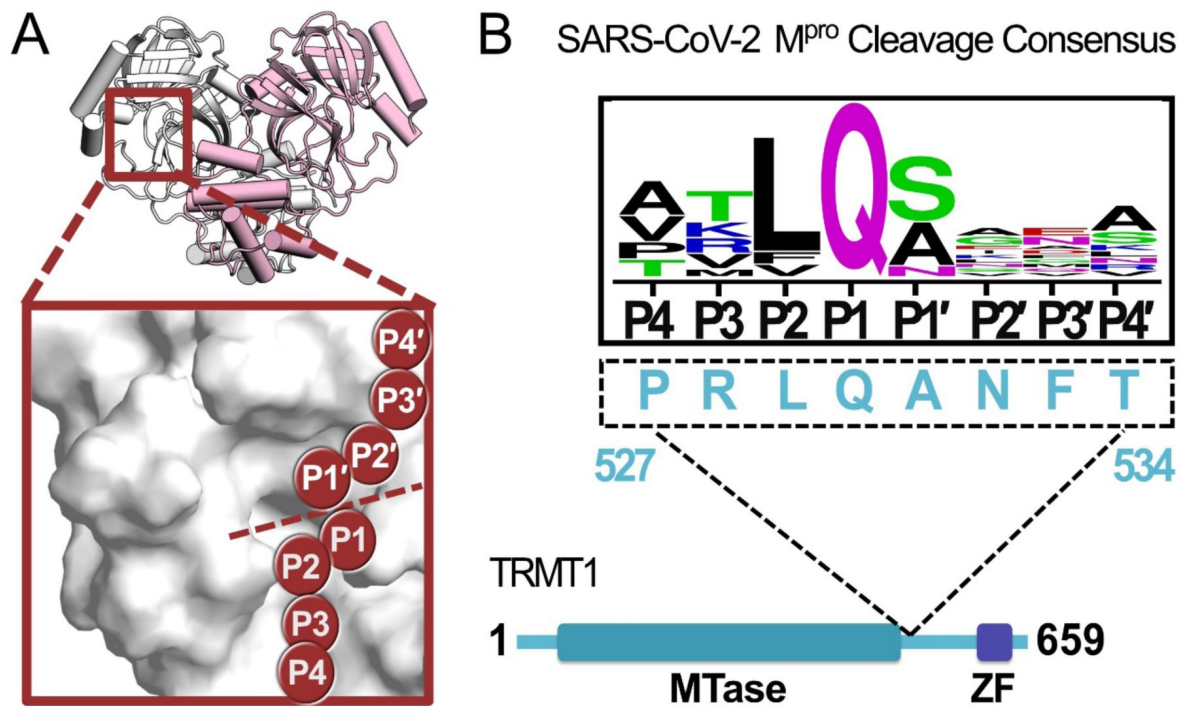


Figure 1.

A peptide sequence found in human TRMT1 fits the cleavage consensus for the SARS-CoV-2 Main Protease (M^{pro}).

A) Overview of the structure of the SARS-CoV-2 M^{pro} homodimer (PDB 7BB2) with substrate peptide residues (P4-P3-P2-P1-P1'-P2'-P3'-P4') illustrated in the M^{pro} active site (inset); proteolytic cleavage takes place between substrate residues P1 and P1' (dotted line). **B)** The TRMT1(527-534) sequence found in a linker region between the TRMT1 SAM methyltransferase (MTase) and Zinc Finger (ZF) domains is consistent with the SARS-CoV-2 M^{pro} cleavage consensus sequence.

were found to have hypersensitivity to redox stress, implicating TRMT1 and the m²,2G²⁶ modification in the regulation of redox homeostasis (21 [↗](#)). TRMT1 contains a peptide sequence (527–534) consistent with the cleavage consensus for M^{Pro}, located in an AlphaFold2-predicted (22 [↗](#), 23 [↗](#)) surface-exposed region (Figure S1) linking the TRMT1 N-terminal SAM-dependent methyltransferase domain and the C-terminal zinc finger domain (**Figure 1B** [↗](#)). SARS-CoV-2 M^{Pro}-directed cleavage of TRMT1, and subsequent downregulation of tRNA m²,2G²⁶ modification during infection, could therefore have direct impacts on both host and viral protein synthesis, as well as phenotypes linked to redox stress.

Here, we show that SARS-CoV-2 M^{Pro} proteolyzes endogenous human TRMT1 in human cell lysate and cleaves an amino acid sequence located between methyltransferase and zinc finger domains in TRMT1 with similar kinetic parameters to known viral polypeptide cleavage sites. M^{Pro}-mediated cleavage of TRMT1 results in reduced tRNA binding affinity and the complete loss of its tRNA methyltransferase activity *in vitro*. We have determined the structure of M^{Pro} in complex with a TRMT1 peptide that shows how the viral protease recognizes and cleaves this human protein sequence and reveals a unique binding mode for the TRMT1 peptide in the M^{Pro} active site. Further kinetic analysis and molecular dynamics simulations show how the flexible M^{Pro} active site can accommodate diverse peptide geometries and suggest that kinetic discrimination between substrate sequences occurs during later steps in the M^{Pro}-catalyzed cleavage reaction. Like many viruses, SARS-CoV-2 hijacks and disrupts diverse host biochemical pathways thereby promoting viral replication and avoiding host detection and immunological response. Recent studies have identified a large number of human proteins that may be targeted and cleaved by M^{Pro}, many of which are involved in regulation of cellular proliferation, inflammatory response, transcription, translation, ubiquitination, apoptosis, and metabolism (24 [↗](#)–30 [↗](#)). Our biochemical and structural data show that TRMT1 can be targeted and cleaved by the SARS-CoV-2 main protease. SARS-CoVs employ multiple strategies to alter host translation (31 [↗](#), 32 [↗](#)), and cleavage of TRMT1 and corresponding loss of the tRNA m²,2G²⁶ modification may be another mechanism by which the virus is able to disrupt or manipulate regulation of host protein synthesis. Furthermore, it is possible that proteolysis of TRMT1 during SARS-CoV-2 infection could contribute to some cellular pathogenesis observed with COVID-19.

Results

Full-length TRMT1 is cleaved by SARS-CoV-2 M^{Pro} *in vitro*

SARS-CoV-1 and SARS-CoV-2 proteome-wide virus-host protein interaction maps identified a putative interaction between catalytically inactive M^{Pro} (Cys145Ala) and human TRMT1, but no such stable interaction was found using wild-type protease, suggesting M^{Pro} may target and cleave TRMT1 in cells (18 [↗](#), 19 [↗](#)). A peptide sequence (527–534) located between the methyltransferase (MTase) domain and zinc finger (ZF) domain of human TRMT1 fits the cleavage consensus sequence of SARS-CoV-2 M^{Pro} (**Figure 1B** [↗](#)). To determine if full-length (FL) TRMT1 is susceptible to cleavage by M^{Pro}, we first measured proteolysis of human TRMT1 expressed and purified from *E. coli* by recombinant purified SARS-CoV-2 M^{Pro} (**Figure 2A** [↗](#)). TRMT1 was incubated with either catalytically inactive M^{Pro} Cys145Ala or active wild-type M^{Pro}, and TRMT1 cleavage was monitored by Western-blot using two TRMT1-specific polyclonal antibodies: a dual-domain recognizing antibody specific for portions of both the methyltransferase and zinc finger domain (anti-TRMT1 460–659), and a single-domain recognizing antibody specific for only the zinc finger domain (anti-TRMT1 609–659). As expected, FL TRMT1 (~75kDa) was stable during a 2-hour incubation with inactive Cys145Ala M^{Pro}. However, when incubated with wild-type M^{Pro}, the intensity of the band corresponding to FL TRMT1, as measured by both TRMT1 antibodies, was reduced by >90% after 2 hours. In Western blots with cleaved TRMT1, the dual domain-recognizing anti-TRMT1(460–659) antibody showed the appearance of two new lower molecular weight bands, corresponding to the anticipated molecular weights of the M^{Pro}-mediated TRMT1

cleavage products (~61 kDa TRMT1 methyltransferase domain and ~14 kDa TRMT1 zinc finger domain). After demonstrating that recombinant FL TRMT1 isolated from *E. coli* could be cleaved by SARS-CoV-2 M^{Pro}, we next asked whether endogenous TRMT1 from human cells could be similarly cleaved by the viral protease. HEK293T cell lysate was incubated with either Cys145Ala or WT M^{Pro}, and endogenous TRMT1 levels were monitored over time by Western blot (**Figure 2B**). Incubation with Cys145Ala M^{Pro} resulted in no change to the FL TRMT1 band over the course of 2 hours, while incubation with wild-type M^{Pro} resulted in time-dependent proteolysis of FL TRMT1. Unlike in cleavage assays with recombinant TRMT1, no build-up of cleavage products was observed during M^{Pro}-mediated proteolysis of endogenous TRMT1, suggesting instability and rapid degradation of cleaved TRMT1 fragments in human cell lysate. Together, these experiments demonstrate that both recombinant FL TRMT1 and endogenous FL TRMT1 in human cell lysate are viable substrates for cleavage by SARS-CoV-2 M^{Pro}.

Cleavage of TRMT1 results in complete loss of tRNA m2,2G modification activity and reduced tRNA binding *in vitro*

Next, we explored the consequences of M^{Pro}-mediated cleavage of TRMT1 on both tRNA m2,2G modification and tRNA binding. TRMT1 was incubated overnight alone, with WT M^{Pro} (to generate a fully cleaved TRMT1 product), or with M^{Pro} Cys145Ala (no cleavage control). The M^{Pro} cleavage reaction was monitored by Western blot using anti-TRMT1(460-659) antibody to confirm the depletion of FL TRMT1 by M^{Pro} WT (**Figure 2C & D**, top). We then measured the impact of TRMT1 cleavage on its tRNA modifying activity by radiolabel-based methyltransferase assays with S-[methyl-¹⁴C]-adenosyl methionine (¹⁴C-SAM; **Figure 2C**, bottom) and its tRNA binding affinity by electrophoretic mobility shift assays (EMSAs) with full-length human tRNA^{phe} (**Figure 2D**, bottom). Cleaved TRMT1 generated by incubation with M^{Pro} WT exhibits a total loss of measurable tRNA methyltransferase activity, compared to TRMT1 incubated with M^{Pro} Cys145Ala, which retains full activity comparable to TRMT1 alone (**Figure 2C**, Table S2). Similarly, we found that M^{Pro}-cleaved TRMT1 also had reduced affinity for tRNA, whereas TRMT1 incubated with M^{Pro} Cys145Ala had comparable tRNA binding affinity to TRMT1 alone (**Figure 2D**, Table S2). While cleaved TRMT1 still retained some affinity for tRNA, we observed in EMSA experiments that cleaved TRMT1-tRNA complexes migrate at significantly higher apparent molecular weights as compared to uncleaved TRMT1-tRNA complexes (Figure S2). This suggests that the TRMT1 cleavage products *in vitro* may form oligomers that have reduced affinity for tRNA, but based on the methyltransferase activity assays above, these oligomeric TRMT1-tRNA complexes cannot efficiently methylate bound tRNA. These observations are consistent with other studies that have shown truncated TRMT1 constructs lacking the zinc finger domain have reduced tRNA modification activity in cells (21) or *in vitro* (33). Together, our *in vitro* tRNA binding and modification assays directly show that M^{Pro}-mediated cleavage of TRMT1 reduces its substrate tRNA binding affinity and completely incapacitates its tRNA m2,2G modification catalysis.

Structure of TRMT1 peptide bound to the M^{Pro} active site

To visualize how TRMT1 is recognized and cleaved by the viral protease active site, we determined the co-crystal structure of catalytically inactive Cys145Ala mutant M^{Pro} in complex with the TRMT1 peptide sequence corresponding to the expected cleavage site, human TRMT1 residues 526-536 (**Figure 3**, Table S1). The 1.9 Å resolution structure shows clear density for the TRMT1 peptide bound to the active site in one protomer of the M^{Pro} dimer (**Figure 3A**, Figure S3A). As expected based on alignment with the M^{Pro} consensus cleavage sequence, TRMT1 Gln530 occupies the critical S1 subsite pocket in the M^{Pro} active site (**Figure 3A**), positioning the Gln530-Ala531 peptide bond directly adjacent to the His41/Cys145 catalytic dyad for proteolytic cleavage (**Figure 3B**). TRMT1 Leu529 corresponds to substrate position P2 and occupies the hydrophobic M^{Pro} S2 pocket (**Figure 3A**), packing against M^{Pro} residues His41, Met49, Met165, and Gln189 (Figure S3B). P3' substrate residue TRMT1 Phe533 occupies the M^{Pro} S3' pocket (**Figure 3A**), which is composed by M^{Pro} residues His41, Cys44, Met49, and the backbone of Thr45 (Figure S3C). TRMT1

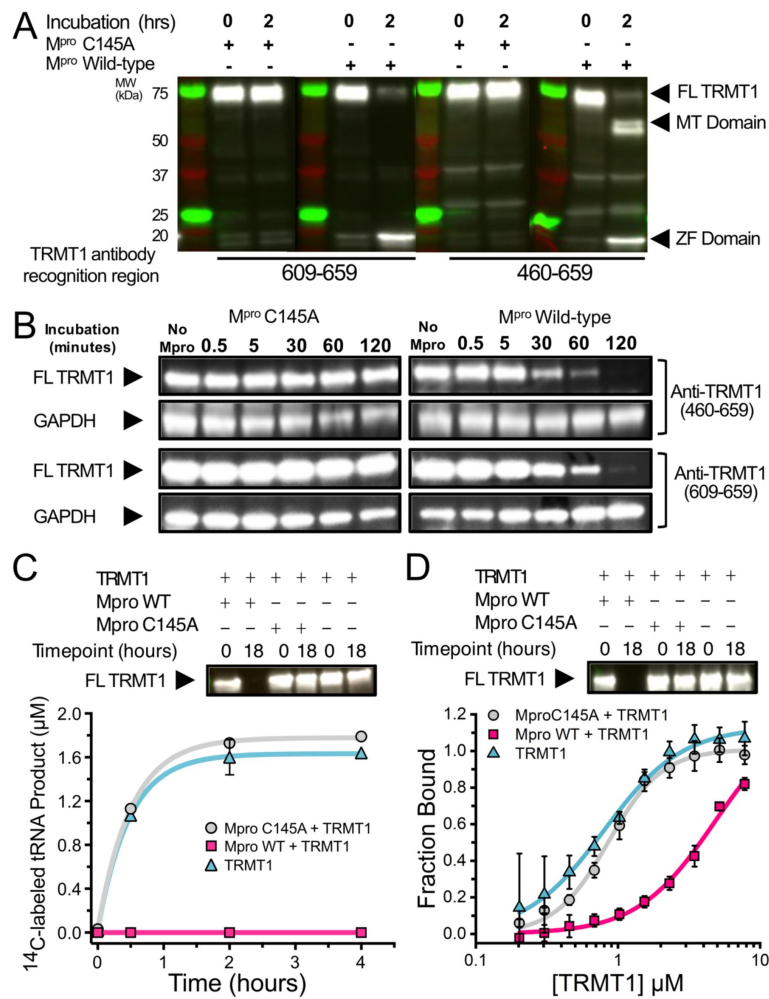


Figure 2.

SARS-CoV-2 M^{pro} cleaves full length human TRMT1 which impacts methyltransferase activity and tRNA binding.

A) Western blots of recombinantly purified full-length TRMT1 incubated with 10 μM catalytically inactive (Cys145Ala) or active (Wild-type) SARS-CoV-2 M^{pro} at 37 °C. Incubation with WT M^{pro} results in proteolysis of FL TRMT1 and the appearance of cleavage products corresponding the ZF domain (observed with both anti-TRMT1(609-659) and anti-TRMT1(460-659) antibodies) and the MTase domain (observed with only anti-TRMT1(460-659) antibody). **B)** Western blots of endogenous human TRMT1 in HEK293T cell lysate incubated with 10 μM of either catalytically inactive (Cys145Ala) or active (Wild-type) M^{pro} at 37 °C. Endogenous FL TRMT1 is stable in human cell lysate over the course of a 2-hour incubation with C145A M^{pro} (left) and is rapidly proteolyzed upon incubation with WT M^{pro} (right). GAPDH was stained in conjunction with TRMT1 antibodies and used as a loading control. **C)** Recombinant, FL TRMT1 cleaved for 18 hours with WT M^{pro} (complete cleavage confirmed by Western blot using anti-TRMT1(460-659), top panel), has no observable methyltransferase activity on a human full-length tRNA^{Phe} substrate (bottom panel, pink squares); in contrast, robust tRNA methylation activity is still seen with TRMT1 incubated for 18 hours with M^{pro} Cys145Ala or no protease (bottom panel, gray circles and blue triangles, respectively). TRMT1 tRNA methyltransferase activity was measured by monitoring radiolabel incorporation into tRNA substrate in reactions with cofactor ¹⁴C-SAM. **D)** Recombinant, FL TRMT1 cleaved for 18 hours with WT M^{pro} (complete cleavage confirmed by Western blot using anti-TRMT1(460-659), top panel), has reduced binding affinity (~6-fold change) for human tRNA^{Phe} (bottom panel, pink squares) compared to uncleaved TRMT1 that had been incubated for 18 hours with either M^{pro} Cys145Ala or no protease (bottom panel, gray circles and blue triangles, respectively). TRMT1-tRNA binding was measured by electrophoretic mobility shift assay (EMSA), where bound and unbound tRNA species at different TRMT1 concentrations were separated, visualized by SYBR Gold staining, and quantified using ImageJ to obtain fraction bound values. Methyltransferase and binding assays in **C** and **D** were carried out in triplicate and errors are shown as SEM; fitted kinetic and binding parameters are shown in Table S2.

peptide binding to M^{Pro} is also mediated by numerous key sidechain and mainchain hydrogen bonding contacts (**Figure 3C, D**). The TRMT1 P1 Gln530 sidechain is specifically recognized by hydrogen bonding interactions with M^{Pro} residues Phe140 (**Figure 3C**) and His163 (**Figure 3D**), consistent with previous structures of M^{Pro} bound to viral peptide substrates (16). Additionally, the TRMT1 P2' Asn532 sidechain interacts with M^{Pro} Asn142 (**Figure 3C**). A number of TRMT1 peptide backbone atoms (from TRMT1 residues Arg528, Leu529, Gln530, Asn532, and Thr534) make hydrogen bond contacts with M^{Pro} surface residues (M^{Pro} Gly143, Glu166, and Gln189 contacts shown in **Figure 3C**; M^{Pro} Thr24, Thr26, Cys145Ala, and His164 contacts shown in **Figure 3D**). The identified M^{Pro}-targeted residues in human TRMT1 are conserved in the human population (*i.e.* no missense polymorphisms), showing that human TRMT1 can be recognized and cleaved by SARS-CoV-2 M^{Pro}.

TRMT1 engages the M^{Pro} active site in a distinct binding conformation

M^{Pro} has more than 10 native viral polypeptide substrates with relatively high sequence variability outside the conserved P1 Gln and the adjacent P2 and P1' residues. We compared our human TRMT1-bound M^{Pro} structure with available SARS-CoV-2 M^{Pro} structures bound to known viral polypeptide cleavage sequences (**Figure 4A**; nsp4/5, nsp5/6, nsp6/7, nsp8/9, nsp9/10, nsp10/11, and nsp15/16). The overall structures of M^{Pro} are very similar for all peptide-bound structures, with M^{Pro} backbone RMSD values all below 1.6 (Figure S4). Analysis of aligned peptide substrates in the M^{Pro} active site shows that both TRMT1 and viral peptide backbones have nearly identical conformations for the N-terminal P4 → P1' residues. In contrast, backbone geometries of C-terminal peptide residues (P2' → P4') diverge more significantly and fall into two distinct binding modes distinguished primarily by the P2' Ψ dihedral angle (**Figure 4B, C**). The majority of viral peptide substrates adopt a binding conformation with P2' $\Psi \geq 157^\circ$ in which the P3' sidechain is positioned away from the M^{Pro} surface (**Figure 4D**), which we have designated as the 'P3'-out' conformation. In contrast, TRMT1 and a single viral peptide substrate, nsp6/7, bind in a distinct conformation with P2' $\Psi \leq 116^\circ$ (**Figure 4C**) in which the P3' sidechain (Phe in TRMT1, Met in nsp6/7) is positioned toward the M^{Pro} surface where it displaces M^{Pro} Met49 to open and occupy pocket S3' (**Figure 4D**). We have designated this uncommon binding mode as the 'P3'-in' conformation.

M^{Pro} cleaves TRMT1 peptide with comparable kinetics to known viral polypeptide cleavage sites

To further examine whether TRMT1 is a viable substrate for SARS-CoV-2 M^{Pro}, we compared kinetic parameters for proteolysis of the TRMT1 (526-536) peptide with the nsp4/5 and nsp8/9 viral polypeptide cleavage sequences, using fluorogenic peptide cleavage assays (Dataset S1). The TRMT1 peptide is cleaved noticeably slower than the nsp4/5 N-terminal auto-processing sequence (**Figure 5A**, Table S3), with an approximately 200-fold decrease in k_{cat} , 4-fold decrease in K_{M} , and 50-fold decrease in catalytic efficiency ($k_{\text{cat}}/K_{\text{M}}$) compared to M^{Pro}-mediated cleavage of the nsp4/5 peptide (**Figure 5B**, Table S3). However, TRMT1 is cleaved with very comparable kinetics to the viral nsp8/9 cleavage sequence, with k_{cat} , K_{M} , and $k_{\text{cat}}/K_{\text{M}}$ values all within ~3-fold for TRMT1 versus nsp8/9 peptide cleavage. These kinetic experiments show that, while the TRMT1 M^{Pro} target sequence is cleaved less efficiently than the canonical nsp4/5 auto-processing site, the TRMT1 sequence is cleaved with kinetics very similar to the known nsp8/9 M^{Pro} cleavage site in the viral polypeptide. This suggests that TRMT1(526-536) is a viable substrate for M^{Pro}-targeted proteolysis.

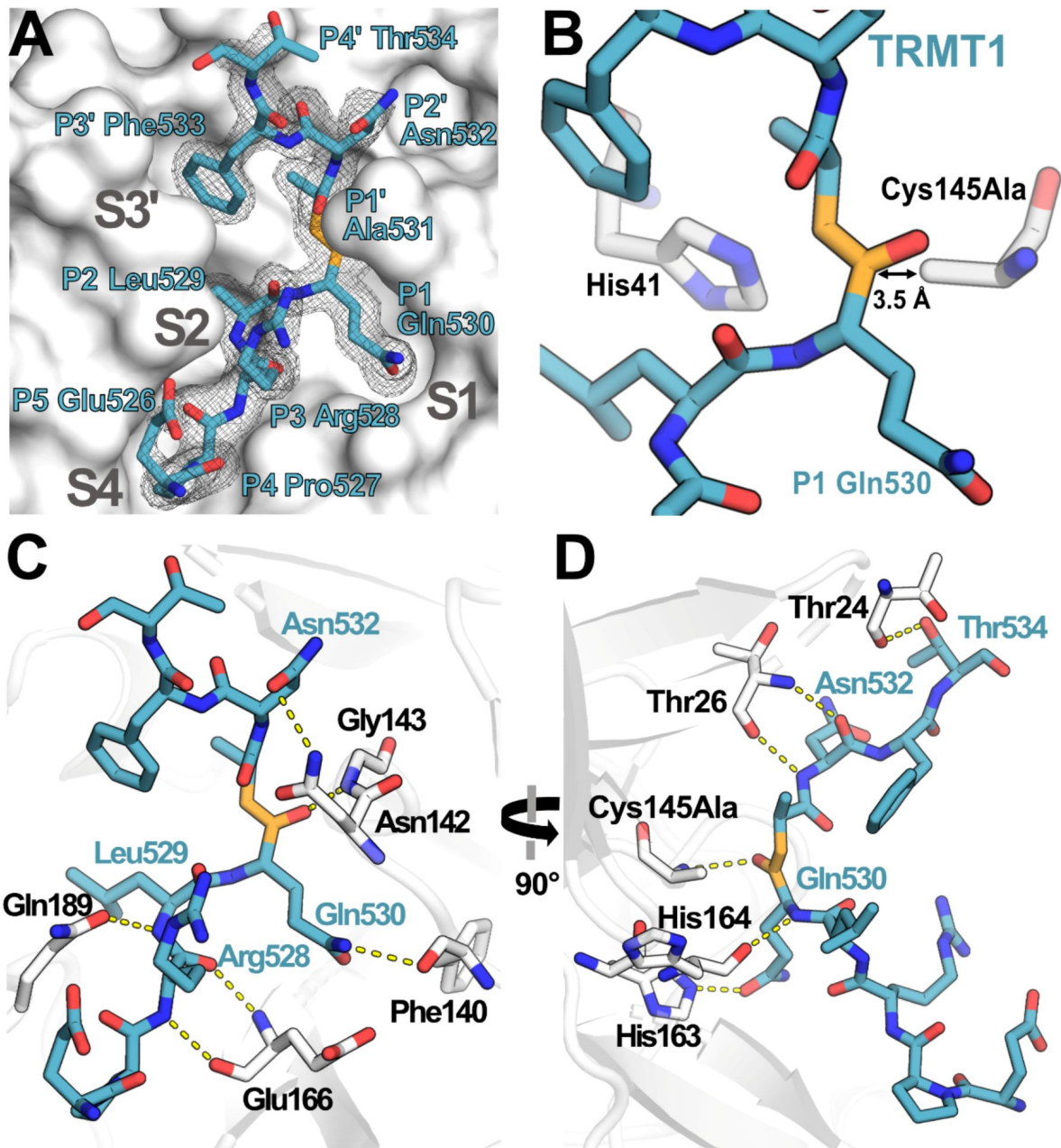


Figure 3.

Structure of human TRMT1(526-536) peptide bound to SARS-CoV-2 M^{Pro} .

Structure of human TRMT1(526-536) peptide bound to SARS-CoV-2 M^{Pro} .

A) TRMT1 peptide (blue) bound in M^{Pro} active site (gray) showing substrate binding pockets S1, S2, S4, and S3'. $F_o - F_c$ omit electron density map of TRMT1 peptide bound to M^{Pro} is shown contoured at 1σ . TRMT1 Gln P1, an ultra-conserved residue in the M^{Pro} cleavage consensus which is critical for M^{Pro} -mediated proteolysis, is nestled in the S1 pocket of the M^{Pro} active site. The scissile P1 – P1' amide linkage of TRMT1 is colored orange. **B**) The TRMT1 P1 Gln amide is positioned for cleavage near M^{Pro} catalytic dyad residues His41 and Cys145Ala in the protease active site. **C & D**) Direct hydrogen bond contacts formed between M^{Pro} residues (white) and the bound TRMT1 peptide (light blue) are illustrated as yellow dashed lines; C and D are views rotated by 90° , highlighting different TRMT1- M^{Pro} hydrogen bonding interactions. M^{Pro} Phe140 and His163 recognize the TRMT1 P1 Gln530 sidechain; TRMT1 Asn532 and M^{Pro} Asn142 engage in sidechain-sidechain interaction; additional backbone hydrogen bond contacts include M^{Pro} Thr24-TRMT1 Thr534, M^{Pro} Thr26-TRMT1 Asn532, M^{Pro} Asn142-TRMT1 Asn532, M^{Pro} Glu166-TRMT1 Arg528, and M^{Pro} Gln189-TRMT1 Leu529; many of these interactions are consistent with canonical M^{Pro} -peptide substrate contacts in the active site.

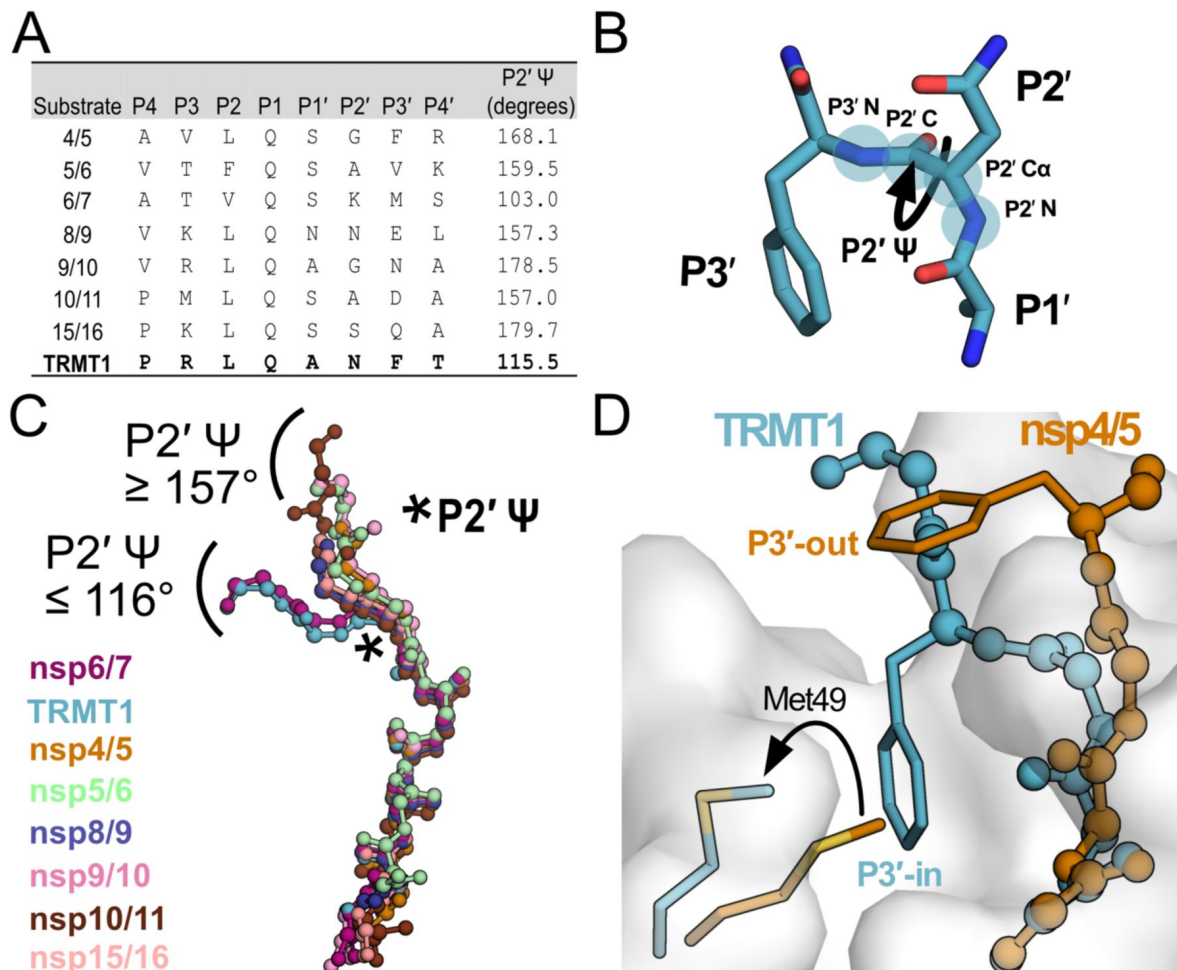


Figure 4.

Analysis of M^{Pro}-peptide structures illustrate two distinct substrate binding modes, P3'-in and P3'-out.

A) Comparison of known M^{Pro} substrate cleavage sequences and the P2' Ψ backbone dihedral angles measured in the corresponding C145A M^{Pro}-peptide structures for each substrate. We included all known C145A M^{Pro}-viral peptide structures in this analysis, except those that were missing the P3' residue or had poorly-defined electron density for the C-terminal portion of the peptide; structures used in this analysis are PDB IDs: 7MGS, 7T8M, 7DVW, 7T9Y, 7TA4, 7TA7, 7TC4, and 9DW6. Additionally, since a C145A M^{Pro}-nsp6/7 structure was not available, we used an H41A M^{Pro}-nsp6/7 structure (PDB 7VDX) for this analysis. **B)** Section of an M^{Pro}-bound peptide substrate showing residues P1', P2', and P3', with the key P2' Ψ dihedral angle illustrated with a curved arrow; the four backbone atoms that define the P2' Ψ dihedral angle are labeled and highlighted with blue circles (P2'N-P2'Ca-P2'C-P3'N). **C)** Alignment of peptide substrate backbones in the M^{Pro} active site reveals two distinct binding modes at the C-terminal end of the bound peptides characterized by P2' Ψ dihedral angles $\geq 157^\circ$ (nsp4/5, nsp5/6, nsp8/9, nsp9/10, nsp10/11, nsp15/16) or $\leq 116^\circ$ (TRMT1, nsp6/7). Peptide overlays were generated by aligning SARS-CoV-2 M^{Pro}-peptide substrate structures in PyMOL. The location of the P2' Ψ dihedral angle in the substrate peptide backbone is denoted with a star. **D)** Alignment of nsp4/5- and TRMT1- bound M^{Pro} structures showing divergent C-terminal peptide substrate binding modes in the M^{Pro} active site. The backbone geometry of nsp4/5 (P2' Ψ = 168°) positions the P3' Phe sidechain away from the M^{Pro} surface ('P3'-out' conformation), while the TRMT1 backbone geometry (P2' Ψ = 115°) positions the P3' Phe sidechain toward the M^{Pro} active ('P3'-in' conformation) site where it displaces M^{Pro} Met49 to open and occupy the S3' pocket.

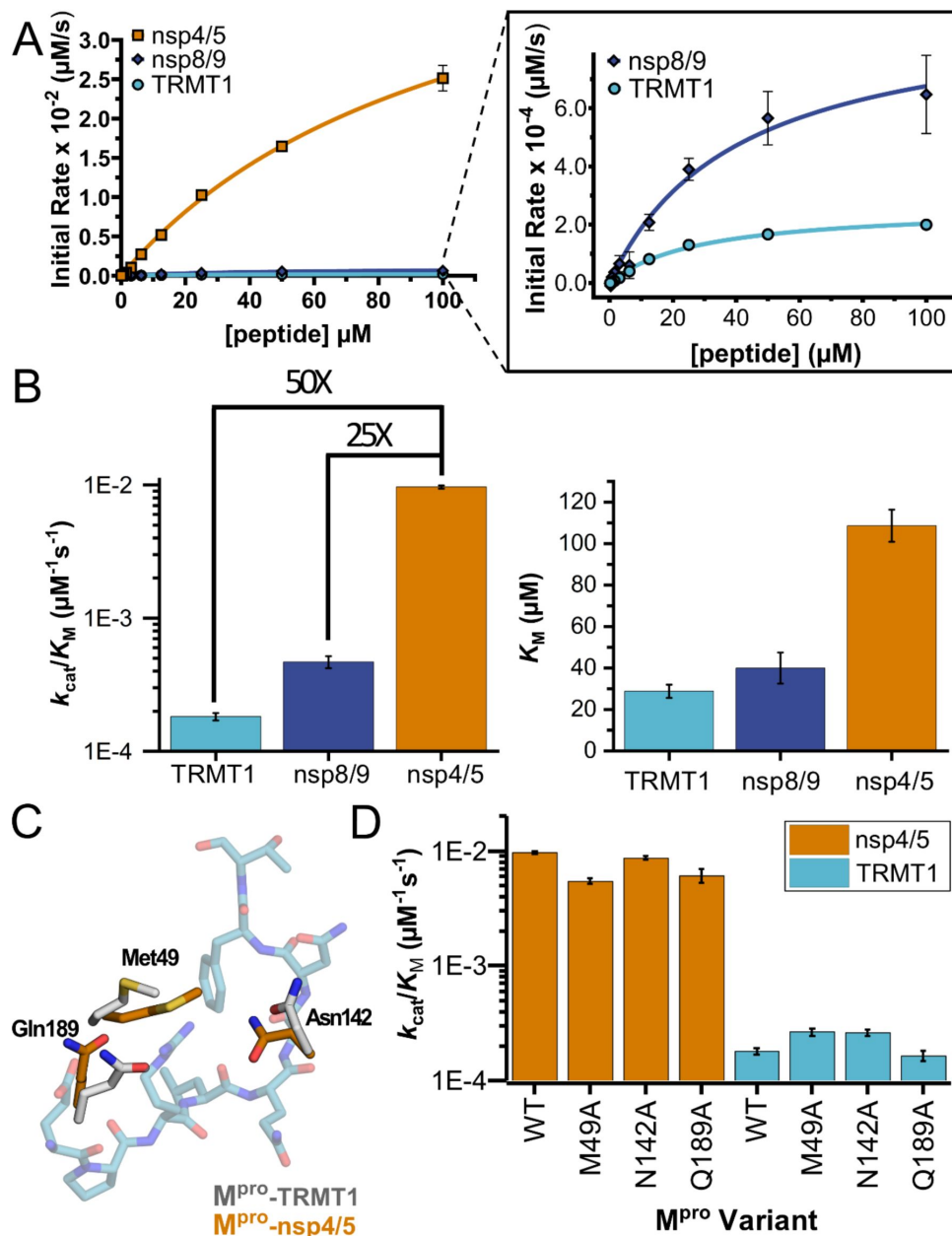


Figure 5.

Human TRMT1 peptides are cleaved with similar catalytic efficiencies to known M^{pro} substrates.

A) Kinetics of nsp4/5, nsp8/9, and TRMT1 peptide cleavage by M^{pro} . To initiate the reaction, 50nM enzyme was added to 100–0.097 μM peptide. Each fluorogenic peptide was conjugated with a quenching moiety, and upon peptide cleavage, the fluorescence of the cleavage product was measured to determine initial rates of the reaction. Nsp4/5 cleavage rates were faster than those observed for the nsp8/9 or TRMT1 peptides, but nsp8/9 and TRMT1 sequences exhibit similar M^{pro} -mediated cleavage rates. **B**) The catalytic efficiency (k_{cat}/K_M) of TRMT1 peptide cleavage by M^{pro} is similar to that for nsp8/9 peptide cleavage; both of these substrates are cleaved significantly slower than the nsp4/5 sequence. This suggests that TRMT1 is a feasible substrate for M^{pro} . **C**) Illustration of changes in M^{pro} Met49, Asn142, and Gln189 residue positioning in TRMT1-bound (white) versus nsp4/5-bound (orange) structures. The TRMT1 peptide is shown in blue; nsp4/5 peptide is not shown. **D**) No major changes in catalytic efficiency are observed for nsp4/5 and TRMT1 peptide cleavage upon mutagenesis of key M^{pro} residues involved in TRMT1 binding and recognition. Primary fluorogenic kinetic data used to construct the plots and determine the kinetic parameters shown in **A** – **D** are listed in Dataset S1; numerical k_{cat} , K_M , and k_{cat}/K_M values are listed in Table S3.

Residues involved in M^{pro}-TRMT1 recognition have only small effects on cleavage efficiency

We next compared the M^{pro}-peptide interactions observed in TRMT1-, nsp4/5-, and nsp8/9-bound structures and identified several M^{pro} residues involved in direct substrate contacts that differ between the TRMT1- and nsp4/5- or nsp8/9-bound structures (**Figure 5C**). In the TRMT1-bound structure, M^{pro} Met49 is shifted to accommodate TRMT1 P3' substrate residue Phe533 binding in the S3' pocket, and the M^{pro} Asn142 and Gln189 sidechains are repositioned to form hydrogen bonds with the TRMT1 peptide backbone. To test how these M^{pro} active site residues affect TRMT1 recognition and catalysis, we generated single alanine point mutations at each of these three sites and measured kinetic parameters for M^{pro}-mediated proteolysis of nsp4/5 and TRMT1 peptides. Additionally, we mutated the P1' TRMT1 Ala531 to Ser, the P1' residue found in nsp4/5 and nearly all of the other M^{pro}-targeted viral cleavage sequences, to test the effects of mutating a key position in the substrate peptide. Surprisingly, we saw no substantial changes in proteolysis kinetics for cleavage of either nsp4/5 or TRMT1 substrates with any of the tested M^{pro} mutants (Met49Ala, Asn142Ala, Gln189Ala) or the TRMT1 Ala531Ser mutant peptide (**Figure 5D**, Table S3, Figure S5), suggesting none of these residues alone play a significant role in determining differential cleavage rates for TRMT1 vs nsp4/5.

Evolutionary insights into TRMT1's conserved M^{pro}

cleavage site reveals M^{pro} proteolytic resistance in rodents

To determine whether the TRMT1 sequence at the M^{pro} cleavage site is unique to human TRMT1 and whether TRMT1 bears signatures of molecular arms-races with pathogens during mammalian evolution, we performed phylogenetic and positive selection analyses (34, 35). Through an evolutionary screen, Cariou *et al.* previously identified some signatures of rapid evolution in primate TRMT1, potentially driven by adaptation to ancient viral pathogens, including coronaviruses (36). Through comprehensive analyses, we identified rapidly evolving sites at the N- and C-termini of primate TRMT1, but we did not find any evidence of sites under positive selection at the M^{pro}-mediated TRMT1(526-536) cleavage sequence (Figure S6A, Dataset S2). In fact, this sequence is highly conserved in primates (**Figure 6A**). A TRMT1 sequence analysis at the mammalian level further showed that the M^{pro} cleavage site of TRMT1 (residues 526-536) is highly conserved in most mammals, including in bats that are the reservoir of SARS-CoVs (**Figure 6B**, Dataset S2). However, one exception is in rodents, where there has been a Q to K substitution fixed in all Muroidea (mouse, rat, hamster...) (**Figure 6C**, S6B, Dataset S2). To test the impact of this substitution on M^{pro}-mediated cleavage, we generated a TRMT1 Gln530Lys mutant and incubated it with M^{pro} WT for a period of two hours (**Figure 6D**). We found that TRMT1 Gln530Lys was resistant to M^{pro} cleavage and did not observe the formation of cleavage products (e.g. TRMT1 MT domain) compared to cleavage of TRMT1 WT under the same proteolytic conditions. This is consistent with a more recent report showing that mouse or hamster TRMT1 is not cleaved by M^{pro} when these enzymes are co-expressed in cells (33). Our results confirm that the TRMT1 Gln530Lys substitution at the invariant P1 glutamine residue is sufficient to completely prevent TRMT1 proteolysis *in vitro*, strongly suggesting this mutation would prevent SARS-CoV M^{pro}-directed TRMT1 cleavage during Muroidea infection.

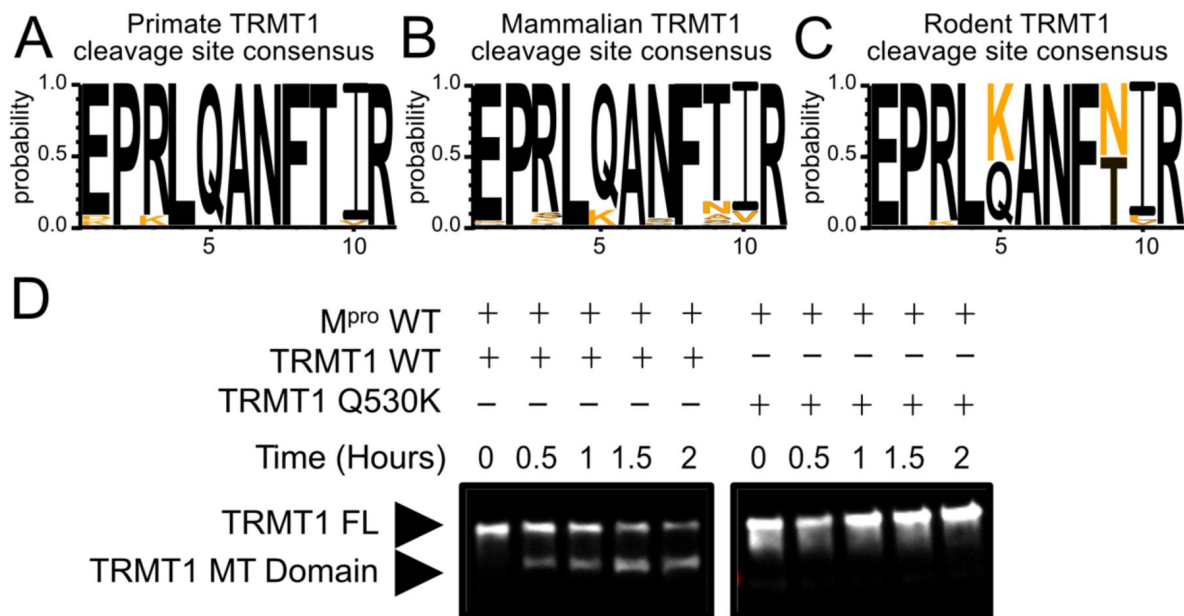


Figure 6.

The M^{pro}-targeted cleavage sequence is conserved in most mammalian TRMT1 proteins, except rodents where TRMT1 is resistant to cleavage.

A – C The M^{pro}-targeted TRMT1 cleavage site sequence (human TRMT1 residues 526-536) is highly conserved in primates (**A**) and most mammals (**B**), with the notable exception of rodents (**C**), where the glutamine Q530 residue most critical for M^{pro}-directed cleavage is substituted to a lysine in Muroidea. Sequence logo plots of the cleavage site in TRMT1(526-536) were produced with WebLogo3. The human reference sequence is in black and orange residues show the differences. **D**) WT human TRMT1 is cleaved over the course of a 2 hour incubation with M^{pro} (left Western blot panel), whereas human TRMT1(Q530K), which has the Q to K mutation found in Muroidea, is entirely resistant to cleavage during a 2 hour incubation with M^{pro} (right Western blot panel).

Molecular dynamics simulations suggest kinetic discrimination happens during later steps of M^{pro}-catalyzed substrate cleavage

To further support the structural and biochemical data described above and attempt to understand how peptide substrate binding geometry may or may not be linked to M^{pro}-catalyzed cleavage efficiency, we carried out molecular dynamics (MD) simulations using nsp4/5-, nsp8/9-, and TRMT1-bound M^{pro} complexes. MD simulations show TRMT1 is stably bound to the M^{pro} active site and primarily adopts the P3'-in conformation observed in the crystal structure, where TRMT1 Phe533 occupies the S3' pocket with a probability of ~65% over the course of the 100 ns simulation ([Figure 7A](#)). In contrast, simulations of the nsp4/5-M^{pro} complex show the nsp4/5 peptide primarily adopts the P3'-out conformation, where nsp4/5 Phe531 is oriented away from the M^{pro} surface with ~70% probability, consistent with the observed nsp4/5 binding mode in the crystal structure. This computational analysis shows that while P3'-in and -out conformations can interconvert, the dominant binding modes for TRMT1 and nsp4/5 peptide substrates (P3'-in vs -out, respectively) are consistent across the crystal structures and MD simulations. Furthermore, because these conformations differ primarily at the C-terminal end of the peptide but have very similar geometries at the scissile amide bond, we find that the computationally and experimentally determined substrate binding poses fail to explain the large differences in cleavage kinetics observed for TRMT1 versus nsp4/5.

Previous structural and geometric analysis of diverse serine protease substrate-inhibitor complexes has shown that the consensus nucleophilic attack angle of the catalytic residue in the reactive Michaelis complex is approximately 90° ([37](#)). We next asked whether differences in nucleophilic attack angle for the catalytic M^{pro} Cys145 measured over the course of the MD simulation might be able to explain changes in M^{pro}-mediated cleavage kinetics. The distribution of the S-C=O nucleophilic attack angle shows that the M^{pro}-nsp4/5 complex has only ~9% increased probability to fall within 89 ± 7°, as compared with the M^{pro}-TRMT1 complex over the course of the simulation ([Figure 7B](#)). Thus, while the nsp4/5 peptide shows slightly more favorable positioning of Cys145 for nucleophilic attack in the Michaelis complex during the first step of cleavage, this is not nearly sufficient to explain the large difference (~200-fold in k_{cat}) in experimentally observed cleavage kinetics between nsp4/5 and TRMT1. Taken together with our TRMT1-M^{pro} structure and kinetic analysis of M^{pro} and TRMT1 mutants, these results strongly suggest that kinetic discrimination between peptide cleavage rates is likely to occur during a later step in the M^{pro}-catalyzed cleavage reaction that follows substrate binding, Michaelis complex formation, and initial nucleophilic attack, similar to some serine proteases ([37](#)).

Discussion

Recent studies mapping host-coronavirus protein interaction networks predicted high confidence interactions between the SARS-CoV-1 or SARS-CoV-2 main protease (M^{pro}) and the human tRNA-modifying enzyme TRMT1 ([18](#), [38](#)). In this work, we show that SARS-CoV-2 M^{pro} can recognize and cleave the human TRMT1 526-536 sequence located between the TRMT1 methyltransferase and zinc finger domains and that cleavage of TRMT1 impairs tRNA substrate binding and abolishes tRNA m2,2G modification activity. M^{pro} is able to efficiently cleave the TRMT1 526-536 sequence in recombinant full-length TRMT1, as well as in endogenous TRMT1 from human cell lysate. The kinetic parameters determined for M^{pro}-mediated TRMT1(526-536) peptide proteolysis are very similar to those measured for the known nsp8/9 viral polypeptide cleavage site. We also determined the structure of human TRMT1(526-536) in complex with SARS-CoV-2 M^{pro} C145A, which reveals a distinct binding mode for the TRMT1 substrate peptide that engages the M^{pro} S3' pocket. Finally, though TRMT1 526-536 is highly conserved in primates and

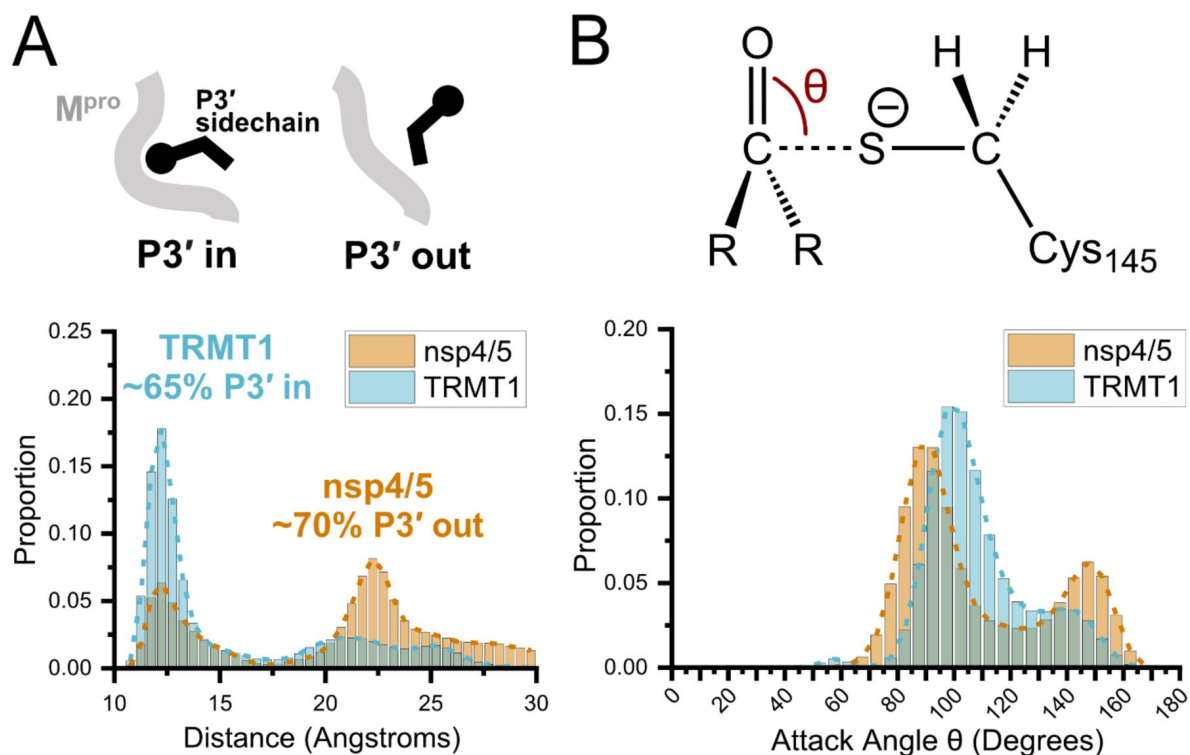


Figure 7.

Molecular dynamics (MD) simulations confirm dominant peptide binding conformations and suggest discrimination in cleavage kinetics result from catalytic steps that follow initial binding and nucleophilic attack.

A) Distribution of the sum of the minimum distance for P3' Phe residue in nsp4/5 or TRMT1 from three residues (Thr25, Met49, Cys44) which form the S3' subsite; P3'-in and P3'-out conformations are illustrated above the distribution plot. The much larger proportion of TRMT1 at smaller distances reflects the peptide's preference for binding in the P3'-in conformation where TRMT1 P3' Phe occupies the S3' pocket during the majority of the MD simulation. **B)** Distribution of the attack angle of the nucleophilic M^{pro} Cys145 sulfur atom and the substrate carbonyl carbon atom in the to-be-cleaved amide bond (S-C=O angle θ , top illustration) during the course of the MD simulation. Although nsp4/5 has a higher proportion of attack angles observed closer to the optimal 90 degrees compared to TRMT1, consistent with faster nsp4/5 cleavage kinetics, this small preference is insufficient to explain the 200-fold faster cleavage kinetics of nsp4/5 observed in experimental proteolysis assays.

mammals, we found that Muroidea contains a critical substitution which confers M^{Pro} cleavage resistance. Our results show that the human tRNA-modifying enzyme TRMT1 is a viable substrate for the SARS-CoV-2 main protease and provide the structural basis for understanding TRMT1 recognition and proteolysis. Importantly, and concurrent with our own work, Zhang *et al.* report that human TRMT1 is cleaved during SARS-CoV-2 infection and that this leads to loss of m²,2G modification in virally infected cells (39). Together, our studies corroborate and characterize cleavage of human TRMT1 by SARS-CoV-2 M^{Pro} from the cellular to the atomic level and raise important new questions about the roles of tRNA methylation during viral infection.

Viruses have evolved diverse strategies that subvert host protein synthesis in order to downregulate host translation and optimize the translation of viral proteins (40–44). SARS-CoV-2 employs multiple mechanisms to disrupt host protein synthesis (12, 45–50), and M^{Pro}-mediated cleavage of human TRMT1 could contribute to the modulation of cellular translation during viral infection. TRMT1 is a tRNA methyltransferase whose m²,2G²⁶ modification activity directly impacts global translational efficiency. Knockouts of TRMT1, or truncations of TRMT1 that remove its zinc finger domain, result in loss of the m²,2G²⁶ tRNA modification and a significant decrease in translation levels in HEK293T cells (21). Our results show that M^{Pro}-mediated cleavage of TRMT1 results in the removal of the zinc finger domain, which weakens TRMT1-tRNA binding affinity and completely abolishes enzyme activity (Figure 2C,D). Additionally, TRMT1-deficient human cells, or cells with inactivating TRMT1 truncations, exhibit increased sensitivity to oxidative stress and links to neurological dysfunction (51–54). Taken together, these observations suggest that M^{Pro}-mediated cleavage of TRMT1 during SARS-CoV-2 infection may contribute to the downregulation of host translation and oxidative stress-related pathogenesis and phenotypes (55). Future experiments in different virally infected cell types will be required to definitively establish how M^{Pro}-directed TRMT1 cleavage impacts cellular and viral translation, viral infectivity and propagation, and oxidative stress phenotypes during infection.

Our evolutionary analyses (Figure 6, S6) raise several possibilities regarding the relationship of SARS-CoV and TRMT1. First, it may be that TRMT1(526–536) is targeted for cleavage by M^{Pro} during coronavirus infection, but that this TRMT1 motif cannot easily mutate during evolution, due to structural constraints or essential function (34, 56–58), and therefore cannot evade viral proteolysis. Indeed, although the TRMT1(526–536) cleavage sequence is found in a linker region between structured domains, an AlphaFold2-predicted (22, 23) TRMT1 structure suggests that many of these residues make contacts with the surface of the TRMT1 MTase domain (Figure S1), which may be important for enzyme stability, domain orientation, or function. Second, it may be that TRMT1 cleavage by coronavirus M^{Pro} exerts insufficient adaptive pressure on the host. In this case, TRMT1 rapid evolution may be driven by other selective pressure or may be due to intrinsically disordered regions. Third, rapidly evolving sites in primate TRMT1 outside of the cleavage site may reflect evasion of SARS-CoV by providing structural escape through within-protein epistasis (59). As coronaviruses infect multiple mammalian species, it would also be informative to our understanding of SARS-CoV evolution and function to further decipher Muroidea TRMT1 resistance to M^{Pro}-mediated cleavage, asking if this resistance confers host advantages or disadvantages during coronavirus infection, and whether this change results from adaptation to ancient coronavirus infection.

Independent of the roles for human TRMT1 and m²,2G RNA modification during SARS-CoV-2 infection, our structural and kinetic results combined with computational analysis highlight distinct features of substrate-protease binding, recognition, and cleavage for SARS-CoV-2 M^{Pro}. First, our TRMT1-M^{Pro} structure shows an uncommon peptide binding mode in which the P3' TRMT1 Phe residue is buried in the S3' pocket on the M^{Pro} surface ('P3'-in' conformation; Figure 4D). Of the currently known M^{Pro}-peptide substrate structures, this unique M^{Pro} P3'-in binding mode is only seen for TRMT1 and nsp6/7, though notably the P3'-in conformation has also been observed in a crystal structure of SARS-CoV-1 M^{Pro} in complex with its C-terminal auto-processing

site (VTFQGKFK), which like TRMT1, contains a phenylalanine at the P3' site (60 [↗](#)). This unique binding mode demonstrates the flexible accommodation of diverse substrate binding poses in the M^{pro} active site and highlights the availability of the S3' pocket for inhibitor binding and therapeutic design. Second, consistent with previous mutagenesis studies of M^{pro} (61 [↗](#)), we find that mutations to residues on the surface of M^{pro} involved in direct binding and recognition of substrate, including Met49, Asn142, and Gln189, have surprisingly little impact on cleavage kinetics (Figure 5D [↗](#)). Similarly, mutating TRMT1 P1' Ala531 to Ser (the P1' residue in nsp4/5 and most other substrate sequences) had no effect on cleavage kinetics and MD simulations still predict this peptide will favor the P3'-in binding conformation (Figure S5). Third, Comparing nsp4/5-, nsp8/9-, and TRMT1-bound M^{pro} crystal structures, the scissile amide bond that links substrate residues P1 and P1' are positioned almost identically in the active site and located at similar distances from the Cys145Ala residue (Figure S7A). Likewise, deviations away from 180 degrees in the dihedral angle of the scissile amide bond in these M^{pro}-peptide structures, which could indicate ground state destabilization and help explain accelerated proteolysis for nsp4/5 (62 [↗](#)), are also inconsistent with the observed trends in peptide cleavage rates (Figure S7B). Molecular dynamics simulations of nsp4/5 and TRMT1 peptide substrates bound to M^{pro} support the observation that the unique P3'-in conformation is favored for TRMT1 binding, but that only subtle differences in geometry are present at the nucleophilic Cys145 residue and scissile peptide bond in the M^{pro}-substrate complexes (Figure 7 [↗](#)), which cannot account for the order of magnitude differences in observed cleavage kinetics between ns4/5 and TRMT1 substrates. Therefore, the large differences in catalytic efficiency observed for M^{pro}-mediated cleavage of nsp4/5 versus nsp8/9 and TRMT1 are not easily explained by structural analysis of the substrate-bound M^{pro} structures or the M^{pro}-peptide geometries and conformations observed in the MD simulations.

Why do a small number of peptide substrate sequences favor the P3'-in conformation? How does substrate sequence or conformation impact cleavage efficiency? Taken together, our structural, kinetic, and computational data described above suggest that: (a) the conformational preferences of bound substrates are not clearly linked to simple sequence features, (b) mutations to individual residues on M^{pro} or its peptide substrate that appear important for M^{pro}-substrate interactions or conformation are often insufficient to change binding conformation or cleavage kinetics, and therefore, (c) kinetic discrimination of substrate cleavage likely happens during later steps of the M^{pro}-mediated proteolysis reaction. Deciphering the underlying, general principles that connect peptide substrate sequence and conformation to M^{pro}-mediated cleavage efficiency remains a significant but important challenge for future structural and computational studies of coronavirus main protease enzymes (63 [↗](#)–68 [↗](#)).

Overall, our work defines the structural basis for understanding recognition and cleavage of the human tRNA methyltransferase TRMT1 by the SARS-CoV-2 main protease and directly shows that TRMT1 cleavage leads to inactivation of its tRNA m2,2G methyltransferase activity. Zhang *et al* provide complementary evidence that TRMT1 is proteolyzed and inactivated during SARS-CoV-2 infection, which would be predicted to affect host and viral translation and increase cellular sensitivity to redox stress (39 [↗](#)). TRMT1 cleavage may therefore have important implications for understanding the ability of SARS-CoV-2 to hijack host protein synthesis and impact cellular phenotypes and COVID-19 pathogenesis. Additionally, our TRMT1-M^{pro} structure highlights a distinct substrate binding mode that reveals the S3' pocket on the surface of the main protease. SARS-Cov-2 M^{pro} is vital for viral propagation and is thus a major target for antiviral drug development (69 [↗](#)–73 [↗](#)). Consideration of the often-hidden S3' pocket on the M^{pro} surface could aid in the design of potent and specific SARS-CoV-2 M^{pro} inhibitors. Currently, the vast majority of inhibitors target the S2 and S1 sites of M^{pro}, with peptidyl inhibitors designed with glutamine and leucine analogs for the P2 and P1 positions, respectively (74 [↗](#)). Our structural results suggest the flexible S3' pocket could be utilized in future therapeutic design, and together with our kinetic analysis adds to the growing understanding of M^{pro} substrate recognition and cleavage efficiency that is critical for potent and specific targeting of M^{pro} to prevent or treat infection.

Materials and methods

Cloning, Protein Expression, and Purification

M^{Pro} constructs used for biochemical and kinetic studies were subcloned into a pETARA expression vector containing N-terminal GST and C-terminal his tags using Gibson assembly. Human codon optimized wild-type M^{Pro} was obtained from Addgene (catalog #141370). M^{Pro} single point mutants (M49A, N142A, Q189A) were introduced using site-directed mutagenesis by whole plasmid PCR. Each construct contained the M^{Pro} autocleavage sequence (AVLQ) after the N-terminal GST tag to allow for self-cleavage and a native M^{Pro} N-terminus. Constructs were transformed into *E. coli* Rosetta™(DE3)pLysS cells and plated on LB agar plates with 50 µg/ml ampicillin. Overnight seed cultures were grown from single colonies in LB media with 50 µg/ml ampicillin, at 37 °C with shaking at 200 rpm. A 1:100 dilution of seed culture to 1 L LB media was prepared and grown at 37 °C with shaking at 200 rpm. After an OD₆₀₀ of 0.6 was reached, cells were induced with 1 mM IPTG and incubated overnight at 18 °C with shaking. Cultures were centrifuged at 7,500 x g and the supernatant was discarded. Harvested cells were sonicated in lysis buffer (25 mM Tris, 300 mM NaCl, 10 mM imidazole pH 8.0), centrifuged at 14,500 x g for 45 minutes, and recovered, clarified lysate was loaded onto 1.5 mL (bed volume) equilibrated Thermo Scientific HisPur Ni-NTA Resin for 30 minutes with gentle mixing. Resin was added to a standard gravity column and washed with two 25 mL washes of lysis buffer and two 25 mL washes with lysis buffer + 25 mM imidazole, and eluted with 10 mL lysis buffer + 250 mM imidazole. Eluate was concentrated to ~2mL using a using 10 kDa MWCO centrifugal concentrator and applied to a Cytiva HiLoad 16/600 Superdex 200 pg column for size exclusion chromatography. M^{Pro}-containing fractions were concentrated to between 10-25 mg/mL, flash frozen with liquid nitrogen, and stored at -70 °C in 50 mM Tris, 1 mM EDTA, 2 mM DTT, pH 7.3.

Catalytically inactive M^{Pro} C145A mutants used for protein crystallography experiments were prepared by subcloning M^{Pro} C145A from Addgene (catalog #141371) into a pET28a expression vector containing an N-terminal His-tag and tobacco etch virus (TEV) protease cleavage site (ENLYFQGS). The M^{Pro} C145A construct was transformed and expressed in *E. coli* as described above. Cultures were centrifuged at 7,500 x g and the supernatant was discarded. Harvested pellets were sonicated in lysis buffer (25 mM Tris, 300 mM NaCl, pH 8.0) and the clarified the lysate by centrifugation at 14,500 x g. Recovered lysate was loaded onto 3.75 mL (bed volume) equilibrated Thermo Scientific HisPur Ni-NTA Resin for 30 minutes with gentle mixing. Resin was added to a standard gravity column and washed with two 25 mL washes of lysis buffer and two 25 mL washes with lysis buffer + 25 mM imidazole, and eluted with lysis buffer + 300 mM imidazole. Eluate was incubated overnight at room temperature with recombinant purified TEV protease at a 1:50 molar ratio to cleave the N-terminal His-tag. A Ni-NTA back pass was performed and the flow-through was concentrated, buffer exchanged into 25 mM Tris, 25 mM NaCl, pH 7.5 using a using 10 kDa MWCO centrifugal concentrator, and applied to a Cytiva HiTrap 5 mL Q HP column. Protein was eluted with a gradient of 25 mM to 1 M NaCl and M^{Pro}-containing fractions were pooled, concentrated and applied to a Cytiva HiLoad 16/600 Superdex 200 pg size exclusion chromatography column and eluted with an isocratic gradient using 25 mM Tris, 25 mM NaCl, pH 7.5. The protein-containing fractions were concentrated to approximately 26 mg/mL, flash frozen with 10% glycerol in liquid nitrogen, and stored at -70 °C.

The human TRMT1 gene sequence (NM_001136035.4) with a 1-18 amino acid N-terminal deletion (to remove the mitochondrial targeting peptide) was synthesized and subcloned by GenScript into pET22b(+) with a N-terminal pelB leader sequence and a C-terminal His-tag. The TRMT1 construct was transformed into *E. coli* Rosetta™(DE3)pLysS cells as described above, and induced using 0.2 mM IPTG. Harvested cells were sonicated in lysis buffer (20 mM Tris-HCl, pH 8.5, 500 mM NaCl, 5 mM imidazole, 10% glycerol, 5 mM BME, cOmplete EDTA-free Protease Inhibitor Cocktail),

clarified at 14,500 x g, and recovered lysate was loaded onto a pre-equilibrated Cytiva 5 mL HisTrap column. The loaded HisTrap column washed with lysis buffer + 20 mM Imidazole, and protein was eluted using a gradient elution with lysis buffer + 250 mM imidazole. Fractions containing TRMT1 were pooled, concentrated, and loaded onto a Cytiva HiLoad 16/600 Superdex 200 pg size exclusion chromatography column and eluted with an isocratic gradient using 20 mM Tris pH 8.5, 500 mM NaCl, 2 mM DTT. The protein-containing fractions were concentrated to approximately 30 mg/mL, flash frozen in liquid nitrogen, and stored at -70 °C.

Crystallography and Structure Determination

Purified M^{Pro} C145A was incubated at room temperature for 1 hour with TRMT1(526-536) peptide (EPRLQANFTIR, synthetic peptide obtained from Peptide 2.0) in 25 mM Tris, 25 mM NaCl, pH 7.4 at a 1:3 molar ratio with the final concentrations of 7 mg/mL M^{Pro} C145A and 619 μM TRMT1(526-536). The M^{Pro} and TRMT1 solution was mixed 1:1 with well solutions consisting of 20-21.5% PEG 3350 and 100 mM NaCl in 24-well hanging drop VDX plates with a final drop volume of 1 μL. Initial crystals were harvested, crushed, and seeded into new hanging drops, with components as listed above, using cat whiskers. Single crystals were harvested and flash frozen in liquid nitrogen with 20% glycerol in well solution as a cryoprotectant. Diffraction data were collected at the National Synchrotron Light Source II (NSLS II) Highly Automated Macromolecular Crystallography (AMX) (75) beamline 17-ID-1 at the Brookhaven National Laboratory on an Eiger 9M Pixel detector at 100 K and a wavelength of 0.920219 Å. Diffraction data were indexed, integrated, and scaled using XDS. The M^{Pro}-TRMT1 structure was solved in space group P2₁2₁2₁ using the Phaser package in the CCP4 suite and a modified PDB 7MGS as search model. After initial rounds of refinement in PHENIX to model M^{Pro} residues, the TRMT1 peptide residues 526 – 534 were manually built into the F_o-F_c map using COOT. Subsequent rounds of automated refinement and water placement using PHENIX and manual adjustments including modeling sodium ions and glycerol molecules in COOT were used to obtain the final structure.

Peptide Cleavage Kinetic Assays

Fluorogenic assays measuring kinetic parameters for cleavage of peptide substrates were carried out in triplicate in Corning Low Volume 384-well Black Flat Bottom Polystyrene NBS Microplates following a similar procedure as Lee, *et al.* (76). Peptide cleavage reactions were carried out at 50 nM M^{Pro} enzyme in 50 mM Tris pH 7.3, 1 mM EDTA, 2 mM DTT, and 20% DMSO, with quenched fluorescent peptide substrates ranging in concentration from 0.097 – 100 μM. Synthetic peptide substrates were obtained from Peptide 2.0: TRMT1 (MCA-EPRLQANFTIR-K(Dnp)K) or nsp4/5 (MCA-SAVLQSGFRKM-K(Dnp)K), where MCA = 7-Methoxycoumarin-4-acetic acid and Dnp = dinitrophenyl. Using a Tecan Spark microplate reader, the fluorescence intensity was monitored every 10 seconds over a 3 minute time course, with excitation at 320 nm and emission at 405 nm. A calibration curve of MCA-AVLQ product fluorescence intensities from 12 to 0.006 μM was measured and used to generate an RFU to μM conversion factor. A correction for the inner filter effect (IFE_{corr}) was determined using the formula $IFE_{corr} = [\text{fluorescence}_{MCA_product+peptide} - \text{fluorescence}_{peptide}] / \text{fluorescence}_{MCA}$ where $\text{fluorescence}_{MCA_product+peptide}$ = the fluorescence of MCA-AVLQ mixed with quenched peptide substrate, $\text{fluorescence}_{peptide}$ = the fluorescence of the peptide alone, $\text{fluorescence}_{MCA}$ = fluorescence of MCA-AVLQ product alone (Dataset S1) (77). Plots of initial rate (μM/min) versus peptide substrate concentration were fit to the Michaelis-Menten equation to determine V_{max} and K_M kinetic parameters for M^{Pro}-mediated peptide cleavage using Origin 2021 software.

Mammalian Cell Culture

Freshly passaged HEK293T cells were gifted from Prof. Catherine Grimes (University of Delaware). Cells were incubated at 37 °C with 5% CO₂ in growth media (DMEM media with 10% FBS and Penicillin (100 U/mL) / Streptomycin (100 μg/mL)). Growth media was replaced every 3 days, and cells were trypsinized and passaged approximately every 7 days, when cells reached 80-90%

confluency. To passage, growth media was aspirated, and cells were gently washed with prewarmed PBS. PBS was then aspirated and a 1:2.5 mixture of 0.25% Trypsin in HBSS with 0.2 g/L EDTA to PBS was added to adherent cells and incubated at 37°C for 2 minutes to disaggregate. A 3X dilution with growth media was used to inactivate trypsin. The cells were gently mixed to attain a homogenous suspension and diluted 1:20 with growth media and added into a new cell culture plate. For lysis, cells were trypsinized as described above, and the resulting cell suspension was spun down at 1250 RPM for 3 minutes. Cell pellets were washed in ice cold PBS 3 times, and spun at 1250 RPM for 3 minutes, aspirating PBS in between each wash. Lysis was performed on a PBS suspension of cells by performing 3 consecutive freeze/thaw cycles by flash freezing in liquid nitrogen and thawing at 37 °C. Lysate was collected after centrifugation at 13,000 RPM for 10 minutes and stored at -70 °C.

Proteolysis Assay

TRMT1 proteolysis reactions with recombinant TRMT1 isolated from *E. coli* were performed with 10 µM Wild-type or C145A M^{Pro} in 20 mM HEPES, 105 mM NaCl, 20% glycerol, pH 7.0. Recombinant full-length TRMT1 from *E. coli* was diluted 7-fold. Reaction was incubated at 37 °C and time points were quenched by adding to SDS-PAGE sample buffer (50 mM Tris, pH 6.8, 1.0% (w/v) SDS, 10% (v/v) Glycerol, 0.1% (w/v) Bromophenol blue, 0.1 M DTT) and boiled for 5 minutes. TRMT1 levels and fragment sizes at different reaction timepoints were assessed by Western blot.

TRMT1 proteolysis assays using endogenous human TRMT1 from HEK293T lysate were carried out using cell lysate preincubated with 1 mM phenylmethylsulfonyl fluoride (PMSF) prior to reaction to prevent proteolysis by mammalian-specific proteases. M^{Pro} proteolysis reactions were performed with 10 µM Wild-type or C145A M^{Pro} in 25 mM Tris, 25 mM NaCl, 20% glycerol, pH 7.5. Reaction was incubated at 37 °C and time points were quenched by adding to SDS-PAGE sample buffer (50 mM Tris, pH 6.8, 1.0% (w/v) SDS, 10% (v/v) Glycerol, 0.1% (w/v) Bromophenol blue, 0.1 M DTT) and boiled for 5 minutes. TRMT1 levels and fragment sizes at different reaction timepoints were assessed by Western blot.

Western Blot

Quenched samples from the proteolysis assays were loaded (10 µL) onto Bio-Rad TGX 4-15% Polyacrylamide gels and run for 30 minutes at 180 V. Gels were blotted onto PVDF membranes using a Bio-Rad Trans blot Turbo for 7 minutes at 2.5 Amps. The blot was incubated in blocking solution (5% non-fat milk in 1X Tris-Buffer Saline with 0.1% Tween) at room temperature for 1 hour. All antibodies were diluted in blocking solution. TRMT1-specific antibodies corresponding to amino acid regions 460-659 (Invitrogen Rabbit Anti-TRMT1 (ref: PA5-96585)) and 609-659 (Bethyl Laboratories Rabbit Anti-TRMT1 (ref: A304-205A)) and housekeeping protein GAPDH (Invitrogen Rabbit Anti-GAPDH (ref: PA-1987)) were utilized for primary antibody staining at 1:2,000 dilution for TRMT1 antibodies and 1:100,000 for GAPDH antibodies, and were stained by overnight incubation at 4 °C. Extensive washing of blot with 1X Tris-buffered saline with 0.1% Tween were performed after primary antibody staining. Invitrogen Goat anti-Rabbit IgG (H+L) Secondary Antibody, HRP (ref: A16096) was used for secondary staining at a dilution of 1:10,000 for 1 hour at room temperature. Extensive washing of blot with 1X Tris-buffered saline with 0.1% Tween were performed after secondary antibody staining. Clarity™ Western ECL Substrate (ref: 1705060) was added to blot and incubated at room temperature for 5 minutes. Western blots were visualized on a Protein Simple FluorChem R imager.

In vitro transcription of tRNA substrate

A double stranded template of tRNA^{phe} (Homo Sapiens Phe-GAA, sequence with T7 promoter: TAATACGACTCACTATAGCCGAAATAGCTCAGTTGGGAGAGCGTTAGACTGAAGATCTAAAG GTCCCTGGTTCGATCCCGGGTTTCGGCA) was used at a concentration of 150 nM in the transcription mixture of 0.05% Triton X-100, 5 mM DTT, 5 mM rNTPs, 0.1 mg/mL RNA polymerase, 2 U/mL TIPP,

40 mM Tris-HCl, pH 7.5, 50 mM MgCl₂, and 2 mM spermidine. The reaction was incubated at 37 °C for 3 hours, then treated with DNase RQ1 and incubated at 37 °C for 1 hour. The reaction was quenched with 50 mM EDTA and 50% formamide and heated for 5 minutes at 95 °C. The quenched material was loaded onto a large scale 8% gel PAGE denaturing UREA gel and run at 50 mA for ~2.5 hours. Using a handheld UV lamp and a silica plate, the tRNA band was excised, shredded, and nuted overnight at room temperature in 0.6 M NaOAc, pH 6.0, 1 mM EDTA, and 0.01 % SDS. The solution was filtered and extracted using Phenol:Chloroform:IAA, 25:24:1, pH 6.6, centrifuging for 5 minutes at 1500 x g and collecting the aqueous phase, then precipitating with 80% ethanol overnight at -20 °C. The tRNA pellet was air dried at room temperature and brought up in molecular grade water. tRNA was then re-folded by heating at 80 °C for 2 minutes, 60 °C for 2 minutes, and then adding 10 mM MgCl₂ and cooling on ice for 30 minutes. The final material was flash frozen at ~150 µM and stored at -80 °C.

14C radiolabel-based methyltransferase activity assays

Overnight M^{pro} cleavage reactions were prepared with 0.5 µM TRMT1 in 50 mM Tris pH 8.0, 50 mM MgCl₂, 100 µg/µL BSA, 2 mM DTT and added to 20 µM M^{pro} WT, 20 µM M^{pro} C145A, or no protease. Timepoints at 0 hours and 18 hours were taken for each protease condition and assessed for reaction completion by Western blot. The completed M^{pro} WT, M^{pro} C154A, or no protease (mock) cleavage reactions were used directly in methyltransferase activity assays with 0.1 µM final concentration of TRMT1, 5 µM tRNA^{phe}, and 30 µM ¹⁴C SAM, (4.2 mCi/mmol) in 50 mM Tris pH 8.0, 50 mM MgCl₂, 100 µg/µL BSA, 2 mM DTT. Reactions were incubated at 37 °C and 5 µL timepoints were taken at 0, 0.5, 2 and 4 hours and processed using Zymo RNA Clean & Concentrator kits to isolate tRNA from the reaction mixture. tRNA was eluted in 15 µL water and added to 10 mL of Ultima Gold Scintillation Fluid (ref: 6013329). Counts per minute (CPM) were measured using Quantulus Scintillation Counter CH3 channel (Lower level 0, Upper level 156, 1 minute count time). A standard curve using known concentrations and specific activity of ¹⁴C-SAM was produced to convert CPM to concentration of labeled product. Graphs were generated using Origin2023b and fit using a MichaelisMenten equation.

Electrophoretic Mobility Shift Assay (EMSA) for detecting tRNA binding

Overnight M^{pro} cleavage reactions were prepared with 14 µM TRMT1 in 20 mM Tris-HCl, pH 8.5, 300 mM NaCl, 2 mM DTT, 20% glycerol and added to 140 µM M^{pro} WT, 140 µM M^{pro} C145A, or no protease. Timepoints at 0 hours and 18 hours were taken for each protease condition and assessed for reaction completion by Western blot. The completed M^{pro} WT, M^{pro} C154A, or no protease (mock) cleavage reactions were used directly to generate a dilution series of TRMT1 and each dilution was mixed with tRNA^{phe} (final TRMT1 concentrations ranged from 7.78 µM to 0.13 µM; final tRNA^{phe} concentration was 0.2 µM). The TRMT1-tRNA binding reactions were incubated for 1 hour at room temperature, 8 µL of each reaction was loaded onto a 5% TBE non-denaturing gel and run at 50-60V for approximately 2.5 hours at 4°C. Each gel was stained with 1X SYBR Gold in 0.5X TBE for 30 minutes at room temperature and then imaged using a Protein Simple FluorChemQ imager. Band intensities for bound and unbound tRNA were quantified using ImageJ software and used to calculate fraction bound values at each TRMT1 concentration. Graphs were generated using Origin2023b and fit using a Hill Plot.

Molecular Dynamics Simulation

Extensive molecular dynamics simulations in explicit water have been carried out to investigate M^{pro}-nsp4/5 and M^{pro}-TRMT1 complexes starting from their corresponding crystal structures (PDB id: 7MGS ([14](#)) and 8D35), respectively. In original crystal structures, the catalytic Cys145 has been mutated to Ala, M^{pro}-nsp4/5 is in the monomer form, while the M^{pro}-TRMT1 complex is in the dimer form. Since the active form of M^{pro} should be dimer ([78](#)), the structure of the M^{pro}-nsp4/5 (PDB id: 7MGS) was superimposed to the M^{pro}-TRMT1 (PDB id: 8D35) to model the dimer

form of M^{Pro}-nsp4/5, and the catalytic Cys145 in both complexes has been modeled using ‘swapa’ command in ChimeraX (79). The original crystal water molecules and ions within 5 Å of the protein-substrate complex are kept. PDB2PQR (80) was employed to add hydrogen atoms to both complexes at pH 7. Subsequently, the His41 was manually changed to Hip41(positively charged) by adding an additional hydrogen atom to the imidazole ring in ChimeraX, and Cys145 was manually changed to Cym(negatively charged) by removing the hydrogen atom on the thiol group in ChimeraX. Each system was neutralized by adding counterions for Na⁺ and Cl⁻, and solvated in a rectangular water box of TIP3P water molecules with 12 Å buffer.

For each prepared simulation system, 3000 steps of steepest descent plus 2000 steps of conjugate minimization was performed with harmonic restraints using a force constant of 20 kcal·mol⁻¹·Å⁻² applied to all heavy atoms coming from the crystal structure, and then the whole system was minimized without restraints by 7000 steps of steepest descent and 3000 steps of conjugate gradient minimization. After minimization, three independent equilibration and MD replicas were carried out for 100ns with the same starting structural configuration but different initial velocities. The equilibration was conducted in five steps: (i) 50 ps constant volume ensemble(NVT) MD simulation with 10 kcal·mol⁻¹·Å⁻² restraints on all heavy atoms from the crystal, and the whole system was heated from 10 to 300K gradually. (ii) 50 ps isothermal isobaric ensemble (NPT) MD with 10 kcal·mol⁻¹·Å⁻² restraints on all heavy atoms from the crystal at 300 K. (iii) 200 ps NPT MD with 5 kcal·mol⁻¹·Å⁻² restraints on all heavy atoms from the crystal at 300 K. (iv) 200 ps NPT MD with 2 kcal·mol⁻¹·Å⁻² restraints on all heavy atoms from the crystal at 300 K. (v) 200 ps NPT MD with 1 kcal·mol⁻¹·Å⁻² restraints on all heavy atoms from the crystal at 300 K. Finally, production MD simulations were carried out for 100 ns at a constant temperature of 300 K and a constant pressure of 1 atm. Langevin thermostat and Berendsen barostat methods were employed to maintain the temperature and pressure, respectively. The atomic coordinates of the complexes were saved every 1ps to obtain the trajectories for analysis.

All MD simulations were conducted with Amber 20 (81) package using Amber FF 14SB force field (82). AmberTools (81) was utilized for preparing topology and coordinate files for the simulated systems. A cutoff of 10 Å was set for calculating van der Waals interactions, and the particle mesh Ewald (PME) (83) method with a cutoff of 10 Å was employed to treat electrostatic interactions. The SHAKE algorithm (84) was used to constrain covalent bonds to allow the integration time step of 2 fs.

Evolutionary analysis of mammalian TRMT1 orthologs

For the mammalian-wide sequence analyses, the TRMT1 amino acid alignment was retrieved with OrthoMaM (Orthologous Mammalian Markers) v10c (85), using human TRMT1 (ENSG00000104907) as query. The cleavage site was located and the sequence logo of the 526- 536 region was generated using WebLogo3 (<https://weblogo.threeplusone.com/>).

For the primate phylogenetic analyses, TRMT1 orthologous sequences were retrieved using the DGINN pipeline with the human TRMT1 CCDS12293 as query (86). TRMT1 sequences from additional primate species were retrieved using NCBI Blastn (Dataset S2).

For the rodent phylogenetic analyses, the rodentia orthologous protein and mRNA reference sequences to the human TRMT1 (ENSG00000104907) gene were collected from OrthoMaM and ncbi HomoloGene (<https://www.ncbi.nlm.nih.gov/homologene>) (Dataset S2). Because most rodent species lack a reference CDS, we used BLOSUM62 matrix implemented in GeneWise from EMBL-EBI tools (87, 88), to identify the TRMT1 ORFs. Codon alignments of primate and rodent TRMT1s were performed using WebPrank (89) with default settings for primates (trust insertions +F, gap rate= 0.05, gap length= 5, K= 2) and with a gap rate of 0.1 for rodents. Positive selection analyses on the codon alignments were performed with HYPHY/Datamonkey (90, 91), using two branch-site models, BUSTED (version 3.1)(92) and aBSREL (version 2.2) (93), and two site- specific models, MEME (version 2.1.2) (94) and FUBAR (version 2.2) (95).

To identify missense polymorphisms or variants in TRMT1 at minor allele frequency above 0.005 in human population, we mined the dbSNP database <https://www.ncbi.nlm.nih.gov/snp/>.

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Data availability

Coordinates and structure factors were deposited in the Protein Data Bank with accession code 9DW6 (note 9DW6 is a re-refinement with very minor changes to our original deposition, 8D35, and supersedes this entry). Data for sequence analyses are further available in Dataset S2, and <https://doi.org/10.6084/m9.figshare.22004474> and <https://doi.org/10.6084/m9.figshare.22004492>. All other data are available from the corresponding author upon reasonable request.

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Reviewer #1 (Public review):

D'Oliviera et al. have demonstrated cleavage of human TRMT1 by the SARS-CoV-2 main protease in vitro. Following, they solved the structure of Mpro (Nsp5)-C145A bound to TRMT1 substrate peptide, revealing binding conformation distinct from most viral substrates. Overall, this work enhances our understanding of substrate specificity for a key drug target of CoV2. The paper is well-written and the data is clearly presented. It complements the companion article by demonstrating interaction between Mpro and TRMT1, as well as TRMT1 cleavage under isolated conditions in vitro. They show that cleaved TRMT1 has reduced tRNA binding affinity, linking a functional consequence to TRMT1 cleavage by MPro. Importantly, the revelation for flexible substrate binding of Nsp5 is fundamental for understanding Nsp5 as a drug target. Trmt1 cleavage assays by Mpro revealed similar kinetics for TRMT1 cleavage as compared to nsp8/9 viral polyprotein cleavage site. They purify TRMT1-Q350K, in which there is a mutation in the predicted cleavage consensus sequence, and confirm that it is resistant to cleavage by recombinant Mpro. I am unable to comment critically on the structural analyses as it is outside of my expertise. Overall, I think that these findings are important for confirming TRMT1 as a substrate of Mpro, defining substrate binding and cleavage parameters for an important drug target of SARS-CoV-2, and may be of interest to researchers studying RNA modifications.

<https://doi.org/10.7554/eLife.91168.2.sa3>

Reviewer #2 (Public review):

Summary:

The manuscript 'Recognition and Cleavage of Human tRNA Methyltransferase TRMT1 by the SARS-CoV-2 Main Protease' from Angel D'Oliviera et al., uncovers that TRMT1 can be cleaved by SARS-CoV-2 main protease (Mpro) and defines the structural basis of TRMT1 recognition by Mpro. They use both recombinant TRMT1 and Mpro as well as endogenous TRMT1 from HEK293T cell lysates to convincingly show cleavage of TRMT1 by the SARS-CoV-2 protease. Using in vitro assays, the authors demonstrate that TRMT1 cleavage by Mpro blocks its enzymatic activity leading to hypomodification of RNA. To understand how Mpro recognizes TRMT1, they solved a co-crystal structure of Mpro bound to a peptide derived from the predicted cleavage site of TRMT1. This structure revealed important protein-protein interfaces and highlights the importance of the conserved Q530 for cleavage by Mpro. They then compare their structure with previous X-ray crystal structures of Mpro bound to substrate peptides derived from the viral polyprotein and propose the concept of two distinct binding conformations to Mpro: P3'-out and P3'-in conformations (here P3' stands for the third residue downstream of the cleavage site). It remains unknown what is the physiological role of these two binding conformations on Mpro function, but the authors established that Mpro has dramatically different cleavage efficiencies for three distinct substrates. In an effort to rationalize this observation, a series of mutations in Mpro's active site and the substrate peptide were tested but unexpectedly had no significant impact on cleavage efficiency. While molecular dynamic simulations further confirmed the propensity of certain substrates to adopt the P3'-out or P3'-in conformation, it did not provide additional insights into the dramatic differences in cleavage efficiencies between substrates. This led the authors to propose that the discrimination of Mpro for preferred substrates might occur at a later stage of catalysis after binding of the peptide. Overall, this work will be of interest to biologists studying proteases and substrate recognition by enzymes and RNA modifications as well as help efforts to target Mpro with peptide-like drugs.

Strengths:

- The authors' statements are well supported by their data, and they used relevant controls when needed. Indeed, they used the Mpro C145A inactive variant to unambiguously show that the TRMT1 cleavage detected in vitro is solely due to Mpro's activity. Moreover, they used two distinct polyclonal antibodies to probe TRMT1 cleavage.
- They demonstrate the impact of TRMT1 cleavage on RNA modification by quantifying both its activity and binding to RNA.
- Their 1.9 Å crystal structure is of high quality and increases the confidence in the reported protein-protein contacts seen between TRMT1-derived peptide and Mpro.
- Their extensive in vitro kinetic assay was performed in ideal conditions although it is sometimes unclear how many replicates were performed.
- They convincingly show how Mpro cleavage is conserved among most but not all mammalian TRMT1 bringing an interesting evolutionary perspective on virus-host interactions.
- The authors test multiple hypotheses to rationalize the preference of Mpro for certain substrates.
- While this reviewer is not able to comment on the rigor of the MD simulations, the interpretations made by the authors seem reasonable and convincing.
- The concept of two binding conformations (P3'-out or P3'-in) for the substrate in the active site of Mpro is significant and can guide drug design.

Weaknesses:

- The two polyclonal antibodies used by the authors seem to have strong non-specific binding to proteins other than TRMT1 but did not impact the author's conclusions or statements. This is a limitation of the commercially available antibodies for TRMT1.
- Despite the reasonable efforts of the authors, it remains unknown why Mpro shows higher cleavage efficiency for the nsp4/5 sequence compared to TRMT1 or nsp8/9 sequences. This is

a challenging problem that will take substantially more effort by several labs to decipher mechanistically.

- The peptide cleavage kinetic assay used by the authors relies on a peptide labelled with a fluorophore (MCA) on the N-terminus and a quencher (Dpn) on the C-terminus. This design allows high-throughput measurements compatible with plate readers and is a robust and convenient tool. Nevertheless, the authors did not control for the impact of the labels (MCA and Dpn) on the activity of Mpro. While in most cases the introduced fluorophore/quencher do not impact activity, sometimes it can.
- An unanswered question not addressed by the authors is if the peptides undergo conformational changes upon Mpro binding or if they are pre-organized to adopt the P3-out and P3-in conformations. This might require substantially more work outside the scope of this immediate article.

<https://doi.org/10.7554/eLife.91168.2.sa2>

Reviewer #3 (Public review):

Summary:

In this manuscript, the authors have used a combination of enzymatic, crystallographic, and in silico approaches to provide compelling evidence for substrate selectivity of SARS-CoV-2 Mpro for human TRMT1.

Strengths:

In my opinion, the authors came close to achieving their intended aim of demonstrating the structural and biochemical basis of Mpro catalysis and cleavage of human TRMT1 protein. The revised version of the manuscript has addressed most of the questions I had posed in my earlier review.

Weaknesses:

Although several new hypotheses are generated from the Mpro structural data, the manuscript falls a bit short of testing them in functional assays, which would have solidified the conclusions the authors have drawn.

<https://doi.org/10.7554/eLife.91168.2.sa1>

Author response:

The following is the authors' response to the current reviews.

Public Reviews:

Reviewer #1 (Public review):

D'Oliviera et al. have demonstrated cleavage of human TRMT1 by the SARS-CoV-2 main protease in vitro. Following, they solved the structure of Mpro (Nsp5)-C145A bound to TRMT1 substrate peptide, revealing binding conformation distinct from most viral substrates. Overall, this work enhances our understanding of substrate specificity for a key drug target of CoV2. The paper is well-written and the data is clearly presented. It complements the companion article by demonstrating interaction between Mpro and TRMT1, as well as TRMT1 cleavage under isolated conditions in vitro. They show that cleaved TRMT1 has reduced tRNA binding affinity, linking a functional consequence to TRMT1 cleavage by MPro. Importantly, the revelation for flexible substrate binding of

Nsp5 is fundamental for understanding Nsp5 as a drug target. Trmt1 cleavage assays by Mpro revealed similar kinetics for TRMT1 cleavage as compared to nsp8/9 viral polyprotein cleavage site. They purify TRMT1-Q350K, in which there is a mutation in the predicted cleavage consensus sequence, and confirm that it is resistant to cleavage by recombinant Mpro. I am unable to comment critically on the structural analyses as it is outside of my expertise. Overall, I think that these findings are important for confirming TRMT1 as a substrate of Mpro, defining substrate binding and cleavage parameters for an important drug target of SARS-CoV-2, and may be of interest to researchers studying RNA modifications.

We thank the reviewer for their positive assessment and summary of our work in this paper!

Reviewer #2 (Public review):

Summary:

The manuscript 'Recognition and Cleavage of Human tRNA Methyltransferase TRMT1 by the SARS-CoV-2 Main Protease' from Angel D'Oliviera et al., uncovers that TRMT1 can be cleaved by SARS-CoV-2 main protease (Mpro) and defines the structural basis of TRMT1 recognition by Mpro. They use both recombinant TRMT1 and Mpro as well as endogenous TRMT1 from HEK293T cell lysates to convincingly show cleavage of TRMT1 by the SARS-CoV-2 protease. Using in vitro assays, the authors demonstrate that TRMT1 cleavage by Mpro blocks its enzymatic activity leading to hypomodification of RNA. To understand how Mpro recognizes TRMT1, they solved a co-crystal structure of Mpro bound to a peptide derived from the predicted cleavage site of TRMT1. This structure revealed important protein-protein interfaces and highlights the importance of the conserved Q530 for cleavage by Mpro. They then compare their structure with previous X-ray crystal structures of Mpro bound to substrate peptides derived from the viral polyprotein and propose the concept of two distinct binding conformations to Mpro: P3'-out and P3'-in conformations (here P3 stands for the third residue downstream of the cleavage site). It remains unknown what is the physiological role of these two binding conformations on Mpro function, but the authors established that Mpro has dramatically different cleavage efficiencies for three distinct substrates. In an effort to rationalize this observation, a series of mutations in Mpro's active site and the substrate peptide were tested but unexpectedly had no significant impact on cleavage efficiency. While molecular dynamic simulations further confirmed the propensity of certain substrates to adopt the P3'-out or P3'-in conformation, it did not provide additional insights into the dramatic differences in cleavage efficiencies between substrates. This led the authors to propose that the discrimination of Mpro for preferred substrates might occur at a later stage of catalysis after binding of the peptide. Overall, this work will be of interest to biologists studying proteases and substrate recognition by enzymes and RNA modifications as well as help efforts to target Mpro with peptide-like drugs.

We thank the reviewer for this thorough and accurate summary of our work in this manuscript.

Strengths:

- *The authors' statements are well supported by their data, and they used relevant controls when needed. Indeed, they used the Mpro C145A inactive variant to unambiguously show that the TRMT1 cleavage detected in vitro is solely due to Mpro's activity. Moreover, they used two distinct polyclonal antibodies to probe TRMT1 cleavage.*
- *They demonstrate the impact of TRMT1 cleavage on RNA modification by quantifying both its activity and binding to RNA.*

- *Their 1.9 Å crystal structure is of high quality and increases the confidence in the reported protein-protein contacts seen between TRMT1-derived peptide and Mpro.*
- *Their extensive in vitro kinetic assay was performed in ideal conditions although it is sometimes unclear how many replicates were performed.*
- *They convincingly show how Mpro cleavage is conserved among most but not all mammalian TRMT1 bringing an interesting evolutionary perspective on virus-host interactions.*
- *The authors test multiple hypotheses to rationalize the preference of Mpro for certain substrates.*
- *While this reviewer is not able to comment on the rigor of the MD simulations, the interpretations made by the authors seem reasonable and convincing.*
- *The concept of two binding conformations (P3-out or P3-in) for the substrate in the active site of Mpro is significant and can guide drug design.*

We thank the reviewer for these positive assessments of manuscript strengths!

Weaknesses:

- *The two polyclonal antibodies used by the authors seem to have strong non-specific binding to proteins other than TRMT1 but did not impact the author's conclusions or statements. This is a limitation of the commercially available antibodies for TRMT1.*

Yes, there are some levels of non-specific binding for all of the TRMT1 antibodies we have tested (this limitation of commercially available TRMT1 antibodies is also observed and noted by Zhang et al), but we agree that this does not impact the overall conclusions and that by using multiple different antibodies to show the same effects, we can have high confidence in the Western blot analysis and interpretation.

- *Despite the reasonable efforts of the authors, it remains unknown why Mpro shows higher cleavage efficiency for the nsp4/5 sequence compared to TRMT1 or nsp8/9 sequences. This is a challenging problem that will take substantially more effort by several labs to decipher mechanistically.*

True! To our knowledge and despite significant past efforts of many research groups studying similar coronavirus proteases (e.g. SARS-CoV-1 Mpro) a clear understanding of the detailed mechanistic relationship between cleavage sequence and cleavage kinetics remains mostly undefined. This is a great and important problem for mechanistic and computational groups with deep interests in proteases to tackle in the future! To highlight these and similar open questions, we have added a short paragraph to the Discussion section (second from the last paragraph).

- *The peptide cleavage kinetic assay used by the authors relies on a peptide labelled with a fluorophore (MCA) on the N-terminus and a quencher (Dnp) on the C-terminus. This design allows high-throughput measurements compatible with plate readers and is a robust and convenient tool. Nevertheless, the authors did not control for the impact of the labels (MCA and Dnp) on the activity of Mpro. While in most cases the introduced fluorophore/quencher do not impact activity, sometimes it can.*

Yes, we agree that it is possible the MCA and Dnp labels could have effects on the measured cleavage rates. These fluorophore/quencher peptide cleavage assays are the standard assays

used by many labs in the protease field to study diverse proteases and diverse cleavage targets. When other labs have compared cleavage kinetic parameters measured with fluorophore/quencher-based peptide cleavage assays versus HPLC-based peptide cleavage assays, these are often found to be quite similar (e.g. Lee, J., Worrall, L.J., Vuckovic, M. et al. Crystallographic structure of wild-type SARS-CoV-2 main protease acyl-enzyme intermediate with physiological C-terminal autoprocessing site. *Nat Commun* 11, 5877 (2020). <https://doi.org/10.1038/s41467-020-19662-4>), although there are also examples where differences arise. In any case, we agree there could be some effects on the cleavage kinetics introduced by the fluorophore and/or quencher groups. However, our main focus in this paper is to show how a sequence in the human tRNA-modifying enzyme TRMT1 is cleaved by Mpro (and in this revision we have also added new data to show the functional effects of cleavage on TRMT1 activity); it will take significant future work to fully dissect the detailed relationships between peptide sequence, including the quantitative effects of fluorophore/quencher labels, and protease-directed cleavage kinetics. Based on our work in this paper and many past studies of similar proteases, understanding how peptide sequence or conformation relates to cleavage efficiency is a longer-term and very challenging problem that we view as beyond the scope of this work. We have added a brief section elaborating on this in the Discussion.

• *An unanswered question not addressed by the authors is if the peptides undergo conformational changes upon Mpro binding or if they are pre-organized to adopt the P3'-out and P3'-in conformations. This might require substantially more work outside the scope of this immediate article.*

We agree this is unanswered; we considered additional MD experiments to address this, but ultimately decided that since both of these sequences are cleaved in the context of much larger polypeptides (FL TRMT1 or the viral polypeptide), any simple analysis to assess the possibility of pre-organization and relate this preferred binding conformation to cleavage kinetics would be difficult to interpret in a biologically meaningful way. We think this and similar questions about how pre-organization of peptides or amino acid sequences in the polypeptides might influence protease binding and cleavage activity are interesting and important future questions for protease-focused groups in this field.

Reviewer #3 (Public review):

Summary:

In this manuscript, the authors have used a combination of enzymatic, crystallographic, and in silico approaches to provide compelling evidence for substrate selectivity of SARS-CoV-2 Mpro for human TRMT1.

Strengths:

In my opinion, the authors came close to achieving their intended aim of demonstrating the structural and biochemical basis of Mpro catalysis and cleavage of human TRMT1 protein. The revised version of the manuscript has addressed most of the questions I had posed in my earlier review.

We thank the reviewer for their positive assessment of this work, and we are glad to hear the manuscript revisions were helpful in addressing the first round of reviews and questions.

Weaknesses:

Although several new hypotheses are generated from the Mpro structural data, the manuscript falls a bit short of testing them in functional assays, which would have solidified the conclusions the authors have drawn.

Toward showing some of the functional effects of TRMT1 cleavage, in this revised version of the manuscript we have added new data and a new results section ('Cleavage of TRMT1 results in complete loss of tRNA m²,2G modification activity and reduced tRNA binding in vitro') showing that cleavage of TRMT1 results in reduced tRNA binding to TRMT1 (Figure 2D) and the complete loss of TRMT1-mediated tRNA modification activity in vitro (Figure 2C). This complements the in-cell data presented by Zhang et al showing that cleavage of TRMT1 in SARS-CoV-2 infected human cells results in the reduction of m²,2G modification levels. We think these data are a strong addition to this paper that broadens the impacts of our reported results more directly into the RNA modifications field.

In terms of showing the further, downstream biological effects of TRMT1 cleavage and/or the specific impacts of TRMT1 cleavage on SARS-CoV-2 propagation and replication, while we agree further functional assays could absolutely heighten the overall impact, we view the main focus of our paper as showing how TRMT1 is recognized and cleaved by Mpro at the structural level and characterizing the biochemistry of the TRMT1-Mpro interaction and the effects of cleavage on TRMT1 tRNA-modifying activity. Zhang et al present some cellular data suggesting that loss of TRMT1 and/or TRMT1 cleavage during infection is actually detrimental to SARS-CoV-2 replication and infectivity. However, a full understanding of how TRMT1-mediated m²,2G modification of tRNA impacts viral translation, whether TRMT1 plays other roles during the viral life cycle, or whether TRMT1 cleavage (even if not important for viral fitness) contributes to cellular phenotypes during infection, will take a significant amount of future cell biology and virology work to unravel. Indeed, our understanding is that characterizing some of the endogenous cleavage targets for the HIV protease and determining the downstream biological effects and impacts on HIV infection took well over a decade. We hope that the biochemical and structural characterization of the Mpro-TRMT1 interaction presented in our paper will provide the necessary fundamental groundwork and impetus for future virology and cellular biochemistry studies to further investigate the biological roles of TRMT1 cleavage by SARS-CoV-2 Mpro.

The following is the authors' response to the original reviews.

eLife Assessment:

This manuscript provides important structural insights into the recognition and degradation of the host tRNA methyltransferase by SARS-CoV-2 protease nsp5 (Mpro). The data convincingly support the main conclusions of the paper. These results will be of interest to researchers studying structures and substrate recognition and specificity of viral proteases.

We thank the eLife editors and reviewers for handling this manuscript and the overall positive assessment of our work.

In this revised version of the manuscript we have included significant, new experimental data with recombinant purified, catalytically active TRMT1 that directly shows cleavage of TRMT1 reduces its tRNA binding affinity (by gel shift assays) and results in the complete loss of tRNA modifying activity in vitro (by radiolabel-based methyltransferase assays). Because these added experiments provide new information about how Mpro-mediated cleavage specifically impacts TRMT1 tRNA binding and m²,2G modification activity, and thus new information about the functional effects of loss of the TRMT1 Zn finger domain, we would strongly suggest adding that "this work may be of interest to researchers studying RNA modifications", or a similar phrase, in the eLife assessment.

Please find below our point-by-point response to each of the reviewer comments, which outlines additional changes to the manuscript.

Public Reviews:

Reviewer #1 (Public Review):

D'Oliviera et al. have demonstrated cleavage of human TRMT1 by the SARS-CoV-2 main protease in vitro. Following this, they solved the structure of Mpro-C145A bound to TRMT1 substrate peptide, revealing binding conformation distinct from most viral substrates. Overall, this work enhances our understanding of substrate specificity for a key drug target of CoV2. The paper is well-written and the data is clearly presented. It complements the companion article by demonstrating the interaction between Mpro and TRMT1 and TRMT1 cleavage under isolated conditions in vitro. Importantly, the revelation of flexible substrate binding of Nsp5 is fundamental for understanding Nsp5 as a drug target. Trmt1 cleavage assays revealed similar kinetics for TRMT1 cleavage as compared to the nsp8/9 viral polyprotein cleavage site, however, it would have been more rigorous for the authors to independently reproduce the kinetics reported for nsp8/9 using their specific experimental conditions. The finding that murine TRMT1 lacks a conserved consensus sequence is interesting, but is not experimentally tested here and is reported elsewhere. I am unable to comment critically on the structural analyses as it is outside of my expertise. Overall, I think that these findings are important for confirming TRMT1 as a substrate of Mpro and defining substrate binding and cleavage parameters for an important drug target of SARS-CoV-2.

We thank the reviewer for their positive assessment and summary of our work in this paper!

We absolutely agree that comparing to nsp8/9 cleavage kinetics measured in our own hands would be more rigorous here, and we have carried out these measurements in triplicate under the same conditions as were used to measure all the other peptide cleavage kinetics in this manuscript. Figures 5A & B (as well as Table S3 and Dataset S2) have been updated with our new nsp8/9 kinetic data ($k_{cat} = 0.019 \pm 0.002 \text{ s}^{-1}$ and $K_M = 40 \pm 7.5 \mu\text{M}$). As expected, our newly measured nsp8/9 kinetic parameters are very similar to those that we had previously cited from MacDonald et al ($k_{cat} = 0.013 \pm 0.001 \text{ s}^{-1}$, $K_M = 36 \pm 6.0 \mu\text{M}$), and show that Mpro-mediated TRMT1 peptide cleavage has similar proteolysis kinetics to the nsp8/9 viral polypeptide cleavage site.

We have also purified full-length human TRMT1 Q530K, which is the key change in the cleavage consensus sequence that likely makes murine TRMT1 resistant to Mpro-mediated cleavage. In in vitro cleavage assays we find that indeed TRMT1 Q530K is entirely resistant to cleavage by recombinant Mpro and we have added this data to the manuscript in Figure 6D. These findings are consistent with previously cited data from Lu et al, which suggest mouse and hamster TRMT1 are not cleaved in HEK293T cells expressing Mpro.

With the addition of the TRMT1 Q530K mutant data, we decided to move the evolutionary analysis together with this kinetic data to a new section in the Results. We think these additions and changes make the paper stronger and clearer, and thank the reviewer for these suggestions!

Reviewer #2 (Public Review):

Summary:

The manuscript 'Recognition and Cleavage of Human tRNA Methyltransferase TRMT1 by the SARS-CoV-2 Main Protease' from Angel D'Oliviera et al., uncovers that TRMT1 can be cleaved by SARS-CoV-2 main protease (Mpro) and defines the structural basis of TRMT1 recognition by Mpro. They use both recombinant TRMT1 and Mpro as well as endogenous TRMT1 from HEK293T cell lysates to convincingly show cleavage of TRMT1 by

the SARS-CoV-2 protease. To understand how Mpro recognizes TRMT1, they solved a co-crystal structure of Mpro bound to a peptide derived from the predicted cleavage site of TRMT1. This structure revealed important protein-protein interfaces and highlights the importance of the conserved Q530 for cleavage by Mpro. They then compared their structure with previous X-ray crystal structures of Mpro bound to substrate peptides derived from the viral polyprotein and proposed the concept of two distinct binding conformations to Mpro: P3-out and P3-in conformations (here P3 stands for the third residue downstream of the cleavage site). It remains unknown what is the physiological role of these two binding conformations on Mpro function, but the authors established that Mpro has dramatically different cleavage efficiencies for three distinct substrates. In an effort to rationalize this observation, a series of mutations in Mpro's active site and the substrate peptide were tested but unexpectedly had no significant impact on cleavage efficiency. While molecular dynamic simulations further confirmed the propensity of certain substrates to adopt the P3-out or P3-in conformation, they did not provide additional insights into the dramatic differences in cleavage efficiencies between substrates. This led the authors to propose that the discrimination of Mpro for preferred substrates might occur at a later stage of catalysis after binding of the peptide. Overall, this work will be of interest to biologists studying proteases and substrate recognition by enzymes as well as help efforts to target Mpro with peptide-like drugs.

We thank the reviewer for this thorough and accurate summary of our work in this manuscript.

Strengths:

- The authors' statements are well supported by their data, and they used relevant controls when needed. Indeed, they used the Mpro C145A inactive variant to unambiguously show that the TRMT1 cleavage detected in vitro is solely due to Mpro's activity. Moreover, they used two distinct polyclonal antibodies to probe TRMT1 cleavage.
- Their 1.9 Å crystal structure is of high quality and increases the confidence in the reported protein-protein contacts seen between TRMT1-derived peptide and Mpro.
- Their extensive in vitro kinetic assay was performed in ideal conditions although it is unclear how many replicates were performed.
- The authors test multiple hypotheses to rationalize the preference of Mpro for certain substrates.
- While this reviewer is not able to comment on the rigor of the MD simulations, the interpretations made by the authors seem reasonable and convincing.
- The concept of two binding conformations (P3-out or P3-in) for the substrate in the active site of Mpro is significant and can guide drug design.

We thank the reviewer for these positive assessments of manuscript strengths!

Weaknesses:

- While the authors convincingly show that TRMT1 is cleaved by Mpro, the exact cleavage site was never confirmed experimentally. It is most likely that the predicted site is the main cleavage site as proposed by the authors (region 527-534). Nevertheless, in Fig 1C (first lane from the right) there are two bands clearly observed for the cleavage product containing the MT Domain. If the predicted site was the only cleavage site recognized by Mpro, then a single band for the MT domain would be expected. This observation suggests that there might be two cleavage sites for Mpro in TRMT1. Indeed, residues

RFQANP (550-555) in TRMT1 might be a secondary weaker cleavage site for Mpro, which would explain the two observed bands in Fig 1C. A mass spectrometry analysis of the cleaved products would clarify this.

We agree with the reviewer that based on the originally presented data it is possible there could be an additional Mpro-targeted cleavage site in TRMT1 beyond the 527-534 region that we validated through peptide cleavage assays of the TRMT1 526-536 peptide. Because it may be difficult to unambiguously identify and differentiate other putative cleavage sites that are nearby to 527-534 (e.g. the suggested possibility of 550-555) by mass spectrometry, we instead carried out additional in vitro cleavage assays with purified FL TRMT1 Q530K. Mutation of the invariant P1 Gln residue in the cleavage sequence is expected to prevent cleavage at this site, and allow us to probe whether there are other sites in TRMT1 that can be cleaved by Mpro (and if so, more straightforwardly identify them by mass spectrometry). We compared cleavage of purified WT FL TRMT1 and FL TRMT1 Q530K with recombinant Mpro in in vitro cleavage assays and found that TRMT1 Q530K is not cleaved by Mpro over the course of a 2h cleavage reaction. In these experiments, we also saw clear cleavage of WT FL TRMT1 over the course of 2h into only a single detectable band. Together, both of these pieces of data strongly suggest that the 527-534 region is the only Mpro-targeted cleavage site in TRMT1 (if there was an additional cleavage site, we should have seen some amount of cleavage in the Q530K mutant, but we do not). Overall, we feel that the updated WT and Q530K experiments clearly demonstrate that there is only one Mpro-mediated cleavage site in human TRMT1, which also is consistent with experiments in Zhang et al showing that Q530N mutations also block TRMT1 cleavage by co-expressed Mpro in human cells.

The updated WT and Q530K cleavage assays have been added to the manuscript in Figure 6D.

• A control is missing in Fig 1D. Since the authors use western blots to show the gradual degradation of endogenous TRMT1, a control with a protein that does not change in abundance over the course of the measurement is important. This is required to show that the differences in intensity of TRMT1 by western blotting are not due to loading differences etc.

Yes, we agree this is an important control and have repeated these experiments and blotted for TRMT1 and GAPDH as a loading control. The updated Western blots are now shown in Figure 2B, and show the same result as the older data.

• The two polyclonal antibodies used by the authors seem to have strong non-specific binding to proteins other than TRMT1 but did not impact the author's conclusions. This is a limitation of the commercially available antibodies for TRMT1, and unless the authors select a new monoclonal antibody specific to TRMT1 (costly and lengthy process), this limitation seems out of their control.

Yes, there are some levels of non-specific binding for all of the TRMT1 antibodies we have tested (this limitation of commercially available TRMT1 antibodies is also observed and noted by Zhang et al), but we agree that this does not impact the overall conclusions and that by using multiple different antibodies to show the same effects, we can have high confidence in the Western blot analysis and interpretation.

• The recombinantly purified TRMT1 seems to have some non-negligible impurities (extra bands in Fig 1C). This does not impact the conclusions of the authors but might be relevant to readers interested in working with TRMT1 for biochemical, structural, or other purposes.

Yes, our initial isolations of recombinant TRMT1 for the first version of this paper produced smaller amounts of TRMT1 with some impurities; we agree that these do not impact the conclusions of the cleavage experiments. However, since our first submission, we have optimized our purification protocols for TRMT1 and are now able to obtain larger quantities of higher purity recombinant human TRMT1 from bacterial cells and we have used this material for the TRMT1 activity and tRNA binding assays added in this revision; we have also included updates to the expression and purification section for recombinant TRMT1. We hope that these improvements will be helpful to readers interested in working on TRMT1.

• *Despite the reasonable efforts of the authors, it remains unknown why Mpro shows higher cleavage efficiency for the nsp4/5 sequence compared to TRMT1 or nsp8/9 sequences.*

True! To our knowledge and despite significant past efforts of many research groups studying similar coronavirus proteases (e.g. SARS-CoV-1 Mpro) a clear understanding of the detailed mechanistic relationship between cleavage sequence and cleavage kinetics remains mostly undefined. This is a great and important problem for mechanistic and computational groups with deep interests in proteases to tackle in the future! To highlight these and similar open questions, we have added a short paragraph to the Discussion section (second from the last paragraph).

• *The peptide cleavage kinetic assay used by the authors relies on a peptide labelled with a fluorophore (MCA) on the N-terminus and a quencher (Dpn) on the C-terminus. This design allows high-throughput measurements compatible with plate readers and is a robust and convenient tool. Nevertheless, the authors did not control for the impact of the labels (MCA and Dpn) on the activity of Mpro. It is possible that the differences in cleavage efficiencies between peptides are due to unexpected conformational changes in the peptide upon labelling. Moreover, the TRMT1 peptide has an E at the N-terminus and an R at the C-terminus (while the nsp4/5 peptide has an S and M, respectively). It is possible that these two terminal residues form a salt bridge in the TRMT1 peptide that might constrain the conformation of the peptide and thus reduce its accessibility and cleavage by Mpro. Enzymatic assays in the absence of labels and MD simulations with the bona fide peptides (including the labels) used in the kinetic measurements are needed to prove that the cleavage efficiencies are not biased by the fluorescence assay.*

These fluorophore/quencher peptide cleavage assays are the standard assays used by many labs in the protease field to study diverse proteases and diverse cleavage targets. When other labs have compared cleavage kinetic parameters measured with fluorophore/quencher-based peptide cleavage assays versus HPLC-based peptide cleavage assays, these are often found to be quite similar (e.g. Lee, J., Worrall, L.J., Vuckovic, M. et al. Crystallographic structure of wild-type SARS-CoV-2 main protease acyl-enzyme intermediate with physiological C-terminal autoproteolysis site. Nat Commun 11, 5877 (2020). <https://doi.org/10.1038/s41467-020-19662-4>), although there are also examples where differences arise. In any case, we agree there could be some effects on the cleavage kinetics introduced by the fluorophore and/or quencher groups or sequence-specific conformational preferences of the peptides. However, because our main focus in this paper is to show how a sequence in the human tRNA-modifying enzyme TRMT1 is cleaved by Mpro (and in this revision we have also added new data to show the functional effects of cleavage on TRMT1 activity), and the broad focus of our lab is understanding the mechanisms controlling the function and activity of RNA-modifying enzymes, we will leave it to other labs focused more specifically on protease biochemistry to fully dissect the detailed relationships between peptide sequence and conformation to protease-directed cleavage kinetics. As discussed above, based on our work in this paper and many past studies of similar proteases, understanding how sequence relates to cleavage

efficiency is a longer-term and very challenging problem that we view as beyond the scope of this work. As noted above, we have added a brief section explaining this in the Discussion.

• The authors used A431S variant in TRMT1-derived peptide to disrupt the P3'-in conformation. While this reviewer agrees with the rationale behind A431S design, it is important to confirm experimentally that the mutation disrupted the P3'-in conformation in favor of the P3'-out conformer. The authors could use their MD simulations to determine if the TRMT1 A431S variant favors the P3'-out conformation.

Thank you for this suggestion; we agree and have carried out the suggested MD simulations with TRMT1 A531S peptides bound to Mpro. Surprisingly, these simulations suggest that the A531S peptide can still readily adopt the P3'-in conformation by orienting the Ser sidechain in a different way as compared to its positioning in the Mpro-nsp4/5 structure. Since this somewhat changes our interpretation of the results of the A531S kinetic experiments, we have rewritten this section of the manuscript by: (a) removing the 'TRMT1 mutations predicted to alter peptide binding conformation have little effect on cleavage kinetics' section in the Results, (b) instead adding several sentences talking about the A531S mutation to the previous section of the results, and including this mutation as another example of how mutations to either Mpro or TRMT1 residues that might be expected to impact cleavage kinetics do not in fact affect cleavage rates, and finally (c) adding the new MD simulation results to the A531S kinetic data in Figure S5 in the Supporting Information. We thank the reviewer for suggesting this important follow-up simulation!

• An unanswered question not addressed by the authors is if the peptides undergo conformational changes upon Mpro binding or if they are pre-organized to adopt the P3'-out and P3'-in conformations.

We agree this is unanswered; we considered additional MD experiments to address this, but ultimately decided that since both of these sequences are cleaved in the context of much larger polypeptides (FL TRMT1 or the viral polypeptide), any simple analysis to assess the possibility of pre-organization and relate this preferred binding conformation to cleavage kinetics would be difficult to interpret in a biologically meaningful way. We think this and similar questions about how pre-organization of peptides or amino acid sequences in the polypeptides might influence protease binding and cleavage activity are interesting and important future questions for protease-focused groups in this field.

• While the authors describe at great length the hydrogen bonds involved in the substrate recognition by Mpro, they occluded to highlight important stacking interactions in this interface. For instance, Phe533 from TRMT1 stacks with Met49 while L529 from TRMT1 packs against His41 of Mpro. Both hydrogen bonding and stacking interactions seem important for TRMT1-derived peptide recognition by Mpro.

Thank you for these suggestions toward additional structural analysis. We have added a short description of L529 packing in the S2 pocket to the main text and Figure S3B. We have also added a short description of F533 packing in the S3' pocket to the main text and Figure S3C.

Reviewer #3 (Public Review):

Summary:

In this manuscript, the authors have used a combination of enzymatic, crystallographic, and in silico approaches to provide compelling evidence for substrate selectivity of SARS-CoV-2 Mpro for human TRMT1.

Strengths:

In my opinion, the authors came close to achieving their intended aim of demonstrating the structural and biochemical basis of Mpro catalysis and cleavage of human TRMT1 protein. The combination of orthogonal approaches is highly commendable.

We thank the reviewer for their positive assessment of this work!

Weaknesses:

It would have been of high scientific impact if the consequences of TRMT1 cleavage by Mpro on cellular metabolism were provided. Furthermore, assays to investigate the effect of inhibition of this Mpro activity on SARS-CoV-2 propagation and infection would have been extremely useful in providing insights into host- SARS-CoV-2 interactions.

Toward showing some of the consequences of TRMT1 cleavage, in this revised version of the manuscript we have added new data and a new results section ('Cleavage of TRMT1 results in complete loss of tRNA m2,2G modification activity and reduced tRNA binding in vitro') showing that cleavage of TRMT1 results in reduced tRNA binding to TRMT1 (Figure 2D) and the complete loss of TRMT1-mediated tRNA modification activity in vitro (Figure 2C). This complements the in-cell data presented by Zhang et al showing that cleavage of TRMT1 in SARS-CoV-2 infected human cells results in the reduction of m2,2G modification levels. We think these data are a strong addition to this paper that broadens the impacts of our reported results more directly into the RNA modifications field.

In terms of showing the further, downstream biological effects of TRMT1 cleavage and/or the specific impacts of TRMT1 cleavage on SARS-CoV-2 propagation and replication, while we agree this would absolutely heighten the overall impact, we view the main focus of our paper as showing how TRMT1 is recognized and cleaved by Mpro at the structural level and characterizing the biochemistry of the TRMT1-Mpro interaction and the effects of cleavage on TRMT1 tRNA-modifying activity. Zhang et al present some cellular data suggesting that loss of TRMT1 and/or TRMT1 cleavage during infection is actually detrimental to SARS-CoV-2 replication and infectivity. However, a full understanding of how TRMT1-mediated m2,2G modification of tRNA impacts viral translation, whether TRMT1 plays other roles during the viral life cycle, or whether TRMT1 cleavage (even if not important for viral fitness) contributes to cellular phenotypes during infection, will take a significant amount of future cell biology and virology work to unravel. Indeed, our understanding is that characterizing some of the endogenous cleavage targets for the HIV protease and determining the downstream biological effects and impacts on HIV infection took well over a decade. We hope that the biochemical and structural characterization of the Mpro-TRMT1 interaction presented in our paper will provide the necessary fundamental groundwork and impetus for future virology and cellular biochemistry studies to further investigate the biological roles of TRMT1 cleavage by SARS-CoV-2 Mpro.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Please list Mpro alias Nsp5 in the Abstract and Introduction, as this is the nomenclature used in the companion article.

OK, we have made these changes.

Reviewer #2 (Recommendations For The Authors):

In addition to the points mentioned in the public review, this reviewer encourages the authors to address the following points:

• Citation 14 is important for this work since the authors used multiple structures from that earlier study for comparison. Citation 14 seems outdated since it refers to a preprint that has been published since then in Nat Comm. The authors should cite the peer-reviewed work <https://pubmed.ncbi.nlm.nih.gov/35729165/>

Thank you, we have updated this reference.

• The description of the hydrogen bonds is tedious to read. The authors could instead classify them into two groups. Hydrogen bonds between main chain backbones or hydrogen bonds between side chains. For instance, they mention the contact between Mpro Glu166-TRMT1 Arg528. This can lead to confusion that a salt bridge is formed while these two residues interact only via their main chain backbones. Indeed, the side chain of R528 is exposed to the solvent.

OK, we have taken this suggestion and tried to simplify and clarify this portion of the text (along with the accompanying structure Figure 3 showing key hydrogen bonds; see below).

• For Figure 2, please label the residues of the peptide with the TRMT1 numbering. This will help the reader to follow the text while looking at the figure.

OK we have added the TRMT1 numbering to what is now Figure 3A, and labeled key TRMT1 residues in Figures 3B, C, and D.

• Fig 2B is important but crowded. The authors could use two panels to show two different views of this interface.

Thank you for this suggestion, we have split B (now C and D in Figure 3) into two panels, rotated 90 degrees from one another, with each view showing a different subset of TRMT1-Mpro interactions. These updated panels are less crowded, and will hopefully be much clearer to readers.

• For increased clarity, the authors could color P3-out in orange and P3-in teal in Fig 3D.

OK, we have made this change.

• Please proofread the method section. There should be a space between values and their units. For example, 20mM HEPES should be 20 mM HEPES.

Thank you, we have corrected these formatting errors in the methods section of the revised version of the manuscript.

• The authors did not identify the mechanism for the higher efficiency of nsp4/5 cleavage despite testing several mutants and MD simulations. Did the author consider changes in the network of water molecules that might be identified in the MD simulations?

We did look at the positioning of waters in nsp4/5 vs nsp8/9 vs TRMT1 MD simulations. In the nsp4/5 simulation we do see a slightly higher density of water molecules positioned at approximately reasonable attack angles for substrate hydrolysis. If we consider water molecules with an attack angle on the scissile amide of 82 – 96 degrees and an attack distance of 4 Å or closer, the probabilities for these conditions in the simulations are: nsp4/5 – 19%, nsp8/9 – 9%, TRMT1 – 6%. More water positioned at reasonable attack positions for nsp4/5 might be consistent with its higher cleavage efficiency, but: (a) these are relatively small differences in water positioning across these 3 Mpro-substrate simulations that would not be

enough to clearly explain the large differences in observed kinetics, and (b) hydrolysis happens in the later steps of the catalytic cycle, so to accurately capture this we would likely need to simulate reaction intermediates formed after initial attack of the active site Cys.

We very much appreciate the reviewer's enthusiasm in pushing us to understand the mechanistic basis for Mpro-directed cleavage efficiencies, and we would have absolutely loved to figure this out! (As it appears to be a long-standing question in the field!) But as discussed above and in the manuscript, we think that it will take a detailed dissection of different steps in the catalytic cycle to understand where and how this selectivity arises. We will leave it to research groups focused more exclusively on the details of protease biochemistry and simulations of reactive intermediates to take up these significant and long-term challenges!

- *In the PDB deposition, Y154 from chain B should be fixed.*
- *In the PDB deposition, some added glycerols seem to conflict. Although this is not important for the biological work discussed in this study, the authors should check if glycerol 403 in chain A and 402, 403 in chain B are properly modeled. Does the density justify placing a glycerol there?*
- *In the PDB deposition, there are over 51 RSRZ outliers. The authors should double-check if they cannot fix them with additional refinements. While such outliers in poorly defined linkers are understandable, this is unexpected for well-defined regions in the map.*

We have made a number of updates to our PDB deposition to address the above three points. (1) We have reexamined and tweaked the loop region at Y154 chain B; this region of the structure has relatively poorly defined electron density, but we now have a model where Y154 is no longer a Ramachandran outlier. The PDB model is now free of any Ramachandran outliers. (2) We have reexamined each of the modeled glycerol molecules and removed one of these (GOL 402), which had a weaker fit to the electron density. The remaining two glycerols appear to be well-modeled (omit maps leaving out each glycerol show strong Fo-Fc density that clearly looks like a glycerol in shape, adding each glycerol back into the model decreases Rwork and Rfree, and the refined 2Fo-Fc map fits well to the modeled glycerols). (3) We agree there are a large number of RSRZ outliers in this structure. We have reexamined many of these, and come to the same conclusion as for our original deposition: that most of these result from residues where there is clear enough density for placing the backbone into the map, but very poor density for the sidechain. Modeling different sidechain positions for the RSRZ outliers we reexamined did not appreciably improve the model fit or change their RSRZ outlier status. For example, Y154 in chains A and B remain some of the worst RSRZ outliers; while the density for these loop regions is generally not very good, it is clear that the backbone atoms of Y154 can be modeled into the structure, but there is very very weak density for the sidechain. We tried modeling alternative and/or multiple sidechain conformations for Y154, but this did not significantly reduce the size of the RSRZ outlier. In short, while we could remove some of these residues or truncate the sidechain where the sidechain density is very poor to lower the total number of RSRZ outliers, we think the best model is one where we leave these residues built into the structure and accept the higher number of RSRZ outliers. Importantly, none of the significant RSRZ outliers are key residues of biological interest that would affect our interpretation of the structure and/or TRMT1-Mpro biochemistry.

We have deposited a new, re-refined PDB model (9DW6) that incorporates these changes and supersedes our old PDB entry (8D35). We have updated the manuscript with the new PDB ID. We thank the reviewer for these suggestions that improved the overall structural model.

Reviewer #3 (Recommendations For The Authors):

The crystal structure entry in the PDB should mention the Cys-to-Ala substitution in Mpro.

Thank you, we have made this change

Fig 2A and 2B: Can the authors highlight the Gln520-Ala531 peptide bind with a different color, please? It gets lost in panel B.

Yes, we have made significant revisions to what is now Figure 3, and have highlighted the scissile peptide bond atoms in orange in each of these panels. Thank you for this suggestion, we agree it helps readers to orient themselves within the structure.

"Importantly, the identified Mpro-targeted residues in human TRMT1 are conserved in the human population (i.e. no missense polymorphisms), showing that human TRMT1 can be recognized and cleaved by SARS-CoV-2 Mpro." Is TRMT1 prone to a high frequency of missense polymorphisms? If so, then this point makes sense. If not, it is not clear if this really informs on any biologically relevant mechanism.

Given (i) that primate TRMT1 was previously identified under positive selection (i.e. rapid evolution) in an evolutionary screen (Cariou et al PNAS 2022) and (ii) that our study is mostly in vitro, we thought it was important to, first, make sure that this sequence of TRMT1 used in functional assays is not specific to a reference sequence that we tested in vitro, but is actually the sequence of TRMT1 in the human population. Further, we were also looking for whether some variations in the Mpro cleavage site of TRMT1 were possibly present in some humans (could these be linked with severe COVID or susceptibility, for example?).

Overall, this statement aims to anchor our in vitro results to the TRMT1 sequences actually present in humans. However, we agree this does not inform "biologically relevant mechanism". We therefore took out the "Importantly" that was probably misleading.

"TRMT1 engages the Mpro active site in a distinct binding conformation."

This is reported as an observation with little analysis. What is the structural basis of this conformational difference between the bound peptides? Why are the psi angles different? Is there a steric factor that is different between these peptide chains? This section can be substantially improved in detail from its current state.

See our related answer to the next comment below.

"Molecular dynamics simulations suggest kinetic discrimination happens during later steps of Mpro-catalyzed substrate cleavage." This section could have partly addressed my previous comment. It is not clear why there is such a large difference in the psi-angle. With access to several peptide-bound structures, the authors should derive and provide insights into the underlying fundamental principles. After all, this is a major point of discovery in their investigation.

We agree that it is not entirely clear why TRMT1 seems to favor the P3'-in conformation when binding to Mpro. The only other known peptide-bound structure that adopts a similar P2' psi angle is nsp6/7, but there are not clear sequence, steric, or interaction features that distinguish TRMT1 and nsp6/7 from the other 6 peptide-Mpro structures that favor a P3'-out conformation with larger P2' psi angle. In particular, the identity of the P1' and P3' residues, which would probably be expected to have the largest impact on this conformation, have no clear commonality in TRMT1 and nsp6/7 that give hints about why these adopt this unique

conformation. As we describe in the discussion section of the manuscript, and has been observed by many other studies of Mpro, the protease active site is very plastic and able to accommodate a diverse range of sequences surrounding the invariant P1 Gln. Furthermore, while the crystal structures of TRMT1 and other nsp cleavage sequences bound to Mpro show a single peptide conformation in the active site, our MD simulations suggest that both P3'-in and P3'-out type conformations are present in solution for TRMT1, nsp4/5, and nsp8/9, just with different populations. It is very likely that there is a delicate energetic balance between these conformations that may depend subtly on multiple sequence features of the peptide and how they interact with each other and the flexible Mpro active site. As with our replies to questions from Reviewer 2 above about deciphering the underlying principles that connect peptide sequence to cleavage efficiency, we expect that dissecting the detailed links between sequence and binding conformation will be a long-term challenge for mechanistic and biocomputational groups focused on viral protease enzymes; systematic mutation of all residues in the cleavage sequence to multiple different amino acid identities followed by structure determination either experimentally and/or computationally will likely be required to uncover the key sequence or steric properties and interactions that underly and drive favored peptide binding conformations.

To highlight these questions as significant and difficult future challenges toward understanding the fundamental principles underlying SARS-CoV Mpro proteolysis, we have added an additional paragraph (second from the last paragraph) in the discussion section.

This work can be taken to a whole new level if the authors were to provide insights into how TRMT1 degradation by Mpro affects host cell biology and how the inhibition of this activity affects CoV biology.

We certainly agree that showing the biological effects of TRMT1 degradation on host cell biology and/or viral biology could raise the impact of this work. But as discussed in more detail above in our response to the weakness listed in Reviewer 3's public review, we see the main focus of this work as showing the biochemical and structural basis for TRMT1 recognition and cleavage by SARS-CoV-2 Mpro, and directly showing the immediate effects of this cleavage on the TRMT1-tRNA interaction and modification activity. As was the case with other viral proteases, like the HIV-1 protease, understanding the potentially diverse and nuanced downstream biological effects of host protein cleavage and its impacts on cellular phenotypes or viral fitness could take many years of careful cell biology and virology work. We hope that our paper provides the key first steps to viral biology labs taking on this significant but important challenge for TRMT1!

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